Rapporto Analisi sollecitazione

File analizzato:	360° Rotating arm v2.ipt
Versione di Autodesk Inventor:	2026.1 (Build 301233020, 233B)
Data di creazione:	21/09/2026, 12:30
Autore studio:	Nuzzolese
Riepilogo:	

Analisi statica:1

Obiettivo generale e impostazioni:

Obiettivo di progetto	Punto singolo
Tipo studio	Analisi statica
Data ultima modifica	21/09/2026, 12:30
Stato modello	[Primario]
Rileva ed elimina le modalità corpo rigido	No

iProperties

Riepilogo

Autore	nuzzo

Progetto

Numero parte	360° Rotating arm v2
Progettista	nuzzo
Costo preventivato	0,00 €
Data creazione	21/09/2026

Stato

Stato progetto	In corso

Fisiche

Materiale	Generico
Densità	1 g/cm^3
Massa	0,0681645 kg
Area	139,809 cm^2
Volume	68,1645 cm^3
Baricentro	x=0,00706241 cm y=5,73384 cm z=0,0704502 cm

Nota: i valori fisici potrebbero essere diversi da quelli utilizzati per FEA riportati di seguito.

Impostazioni mesh:

Dimensione media elemento (frazione del diametro modello)	0.1
Dimensione minima elemento (frazione della dimensione media)	0.2
Fattore di gradazione	1.5
Angolo di rotazione massimo	60 gr
Crea elementi mesh curvi	Sì

Materiale/i

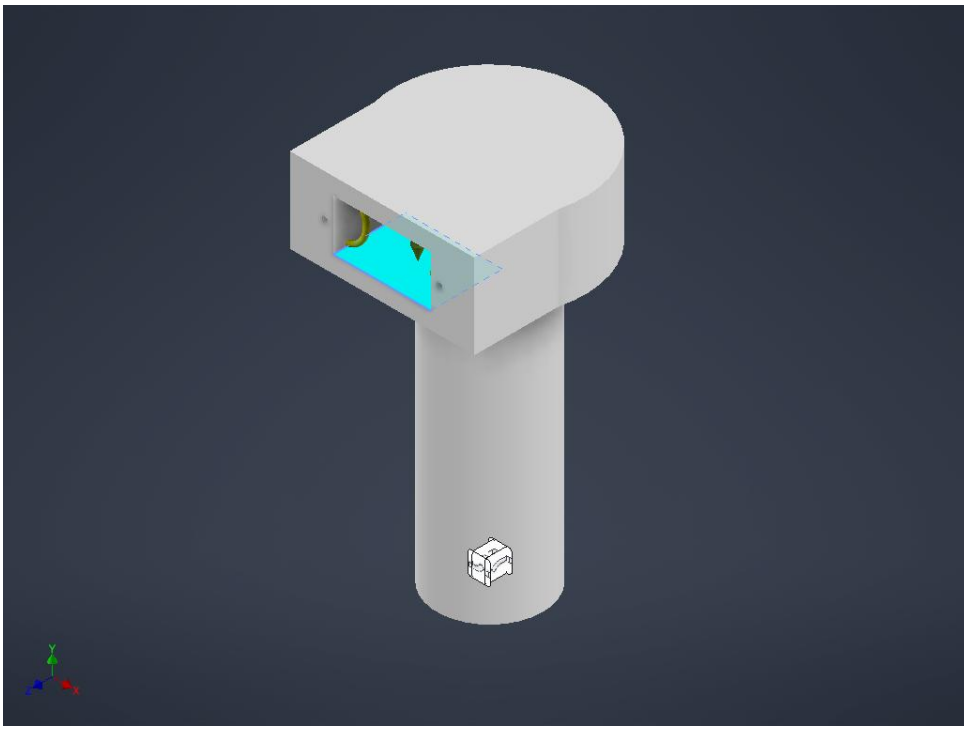
Nome	Plastica ABS	
Generale	Densità della massa	1,06 g/cm^3
	Resistenza allo snervamento	20 MPa
	Resistenza massima a trazione	29,6 MPa
Sollecitazione	Modulo di Young	2,24 GPa
	Coefficiente di Poisson	0,38 su
	Modulo a taglio	0,811594 GPa
Nome/i parte	360° Rotating arm v2.ipt	

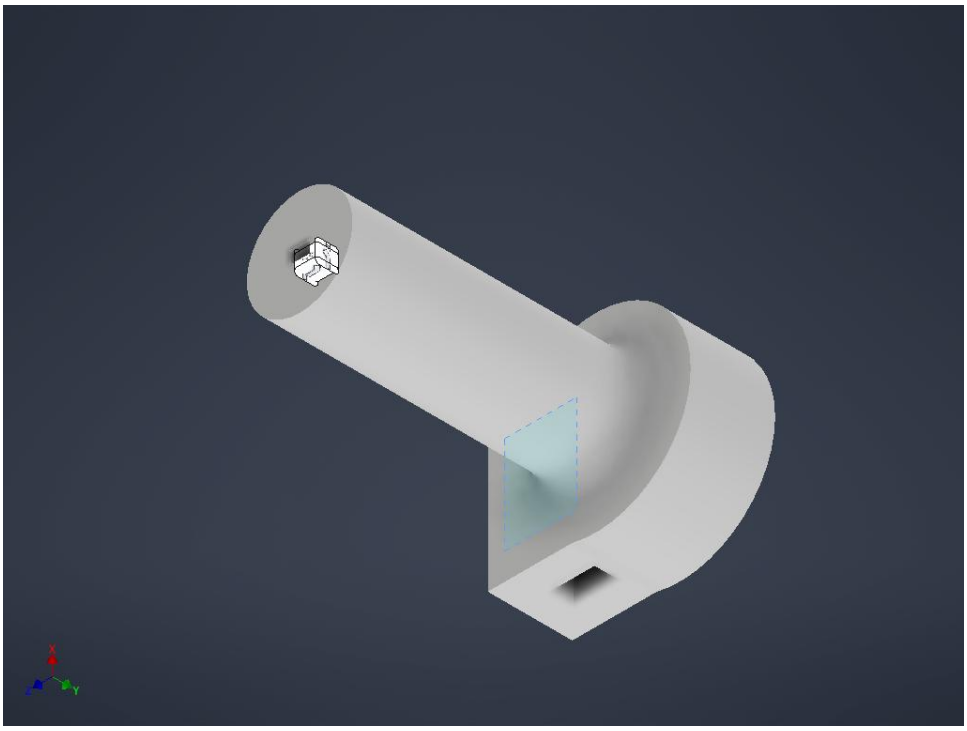
Condizioni operative

Forza:1

Tipo carico	Forza
Intensità	15,000 N
X vettore	0,000 N
Y vettore	-15,000 N
Z vettore	0,000 N

Faccia/e selezionate

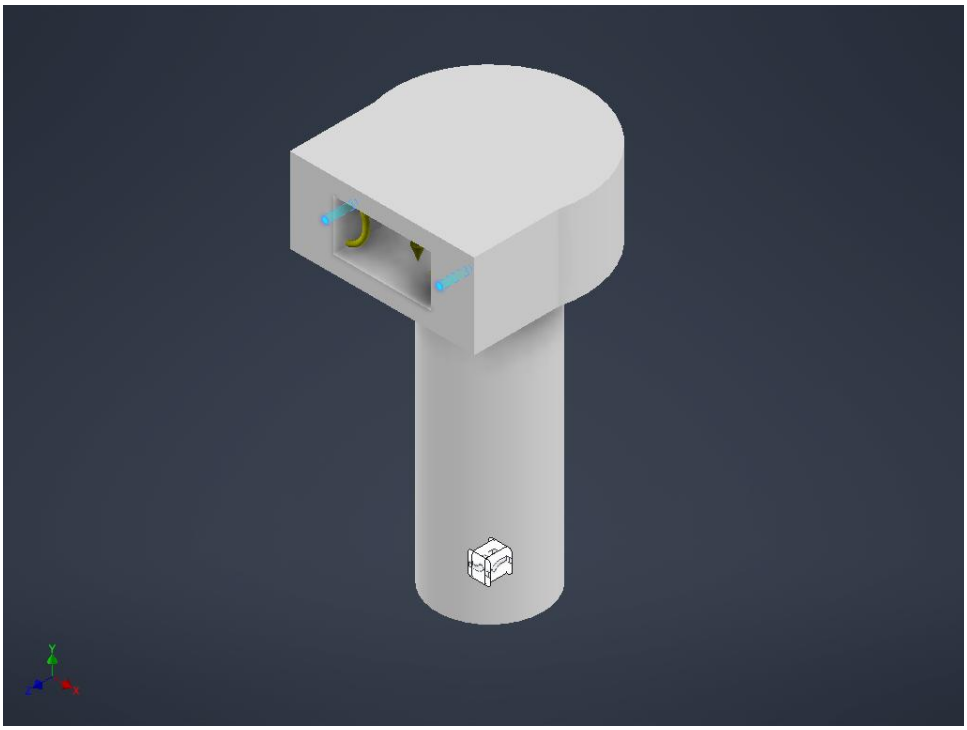


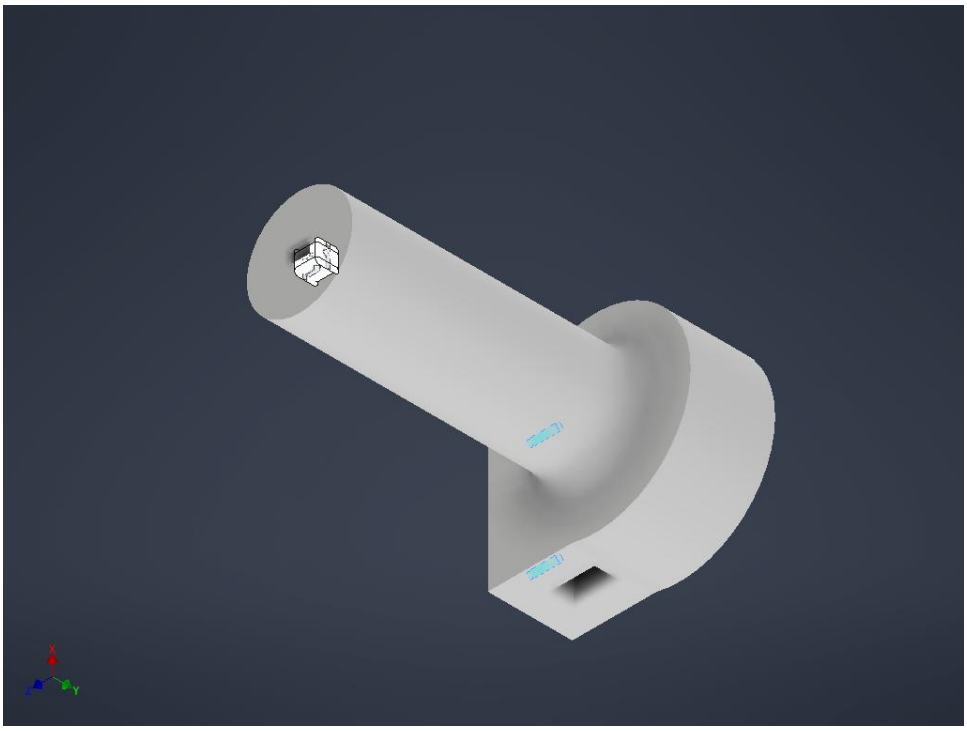


Momento:1

Tipo carico	Momento
Intensità	300,000 N cm
X vettore	0,000 N cm
Y vettore	0,000 N cm
Z vettore	-300,000 N cm

Faccia/e selezionate



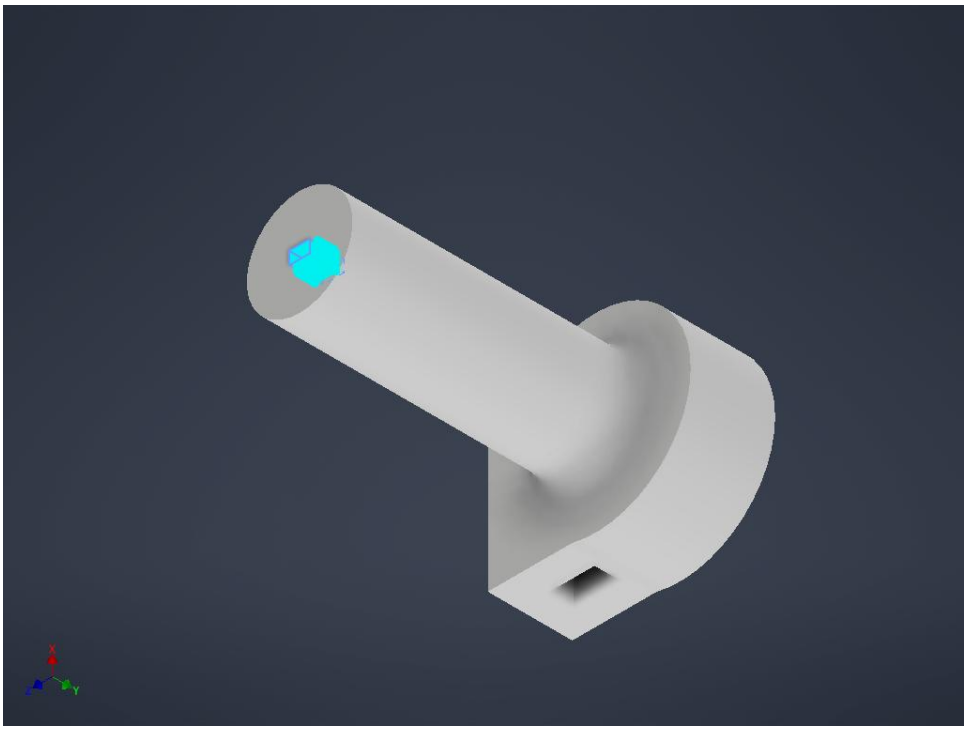


Vincolo di fissaggio:1

Tipo vincolo	Vincolo di fissaggio

Faccia/e selezionate





Risultati

Forza di reazione e momento su vincoli

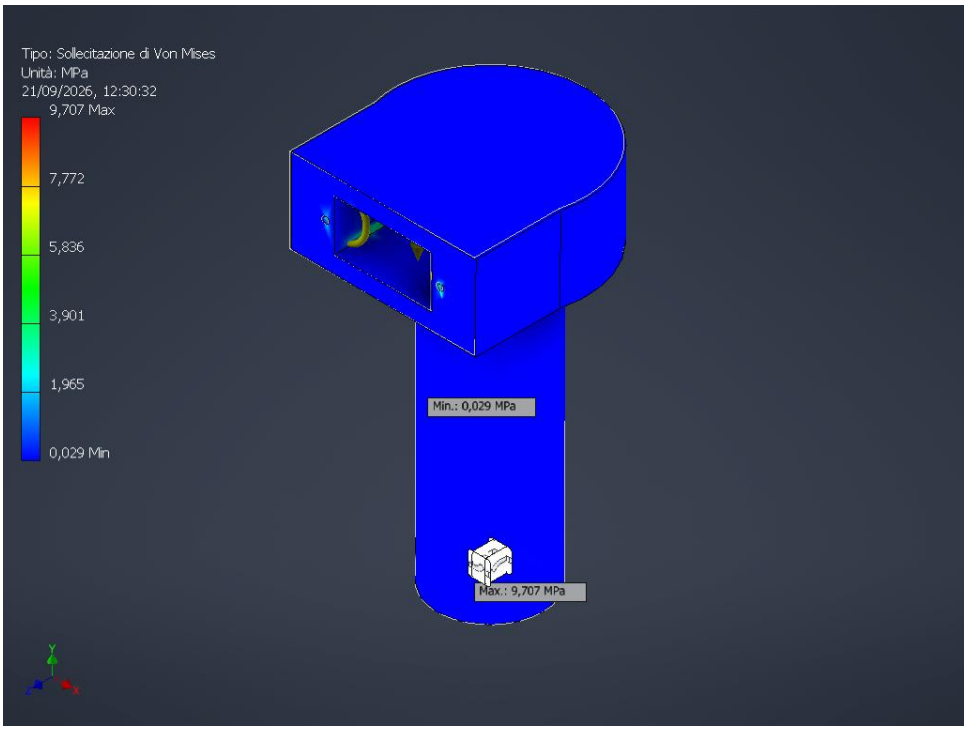
Nome vincolo	Forza di reazione		Momento di reazione	
	Intensità	Componente (X,Y,Z)	Intensità	Componente (X,Y,Z)
Vincolo di fissaggio:1	15,2198 N	0 N	3,01193 N m	-0,260588 N m
		15,2198 N		0 N m
		0 N		3,00064 N m

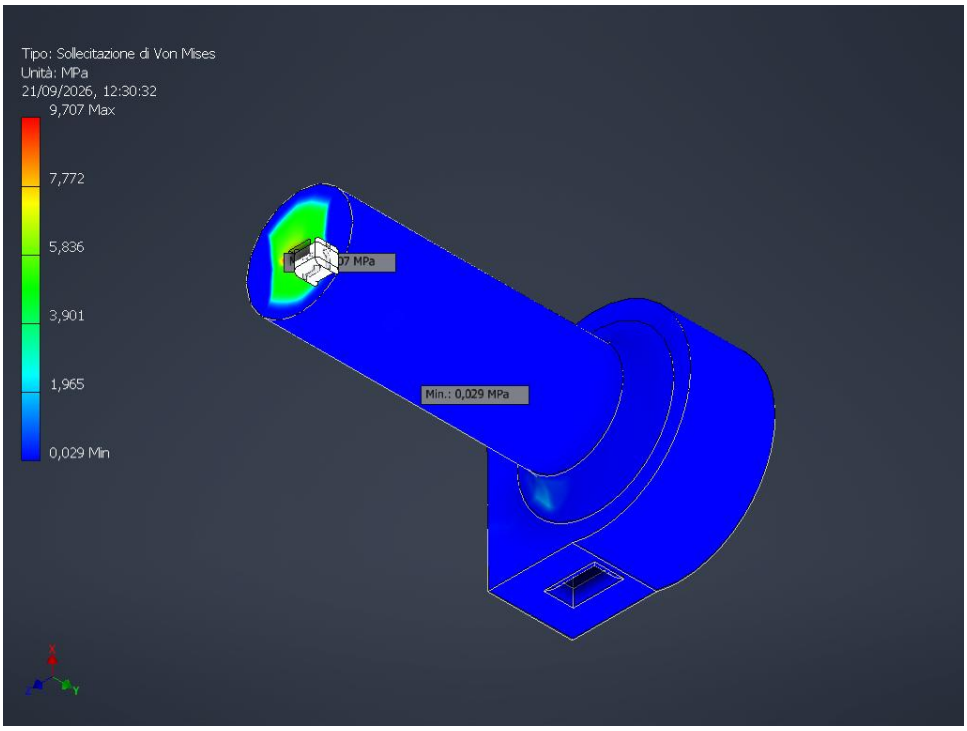
Riepilogo risultati

Nome	Minimo	Massimo
Volume	68164,7 mm^3	
Massa	0,0722546 kg	
Sollecitazione di Von Mises	0,0294613 MPa	9,70712 MPa
Prima sollecitazione principale	-5,71343 MPa	13,7568 MPa
Terza sollecitazione principale	-13,8946 MPa	5,54111 MPa
Spostamento	0 mm	0,524506 mm
Fattore di sicurezza	2,06034 su	15 su
Sollecitazione XX	-12,157 MPa	11,9647 MPa
Sollecitazione XY	-3,38769 MPa	3,5915 MPa
Sollecitazione XZ	-5,314 MPa	4,06939 MPa
Sollecitazione YY	-7,45105 MPa	7,33321 MPa
Sollecitazione YZ	-2,77099 MPa	3,43658 MPa
Sollecitazione ZZ	-7,45105 MPa	7,33321 MPa
Spostamento X	-0,016476 mm	0,493122 mm
Spostamento Y	-0,208399 mm	0,178636 mm
Spostamento Z	-0,00292046 mm	0,0583608 mm
Deformazione equivalente	0,0000127599 su	0,00399012 su
Prima deformazione principale	0,0000063947 su	0,00395739 su
Terza deformazione principale	-0,00396969 su	-0,0000100857 su
Deformazione XX	-0,00289919 su	0,00285334 su
Deformazione XY	-0,00208706 su	0,00221262 su
Deformazione XZ	-0,0032738 su	0,00250703 su
Deformazione YY	-0,00184141 su	0,00178281 su
Deformazione YZ	-0,00170713 su	0,00211718 su
Deformazione ZZ	-0,00156007 su	0,00136786 su

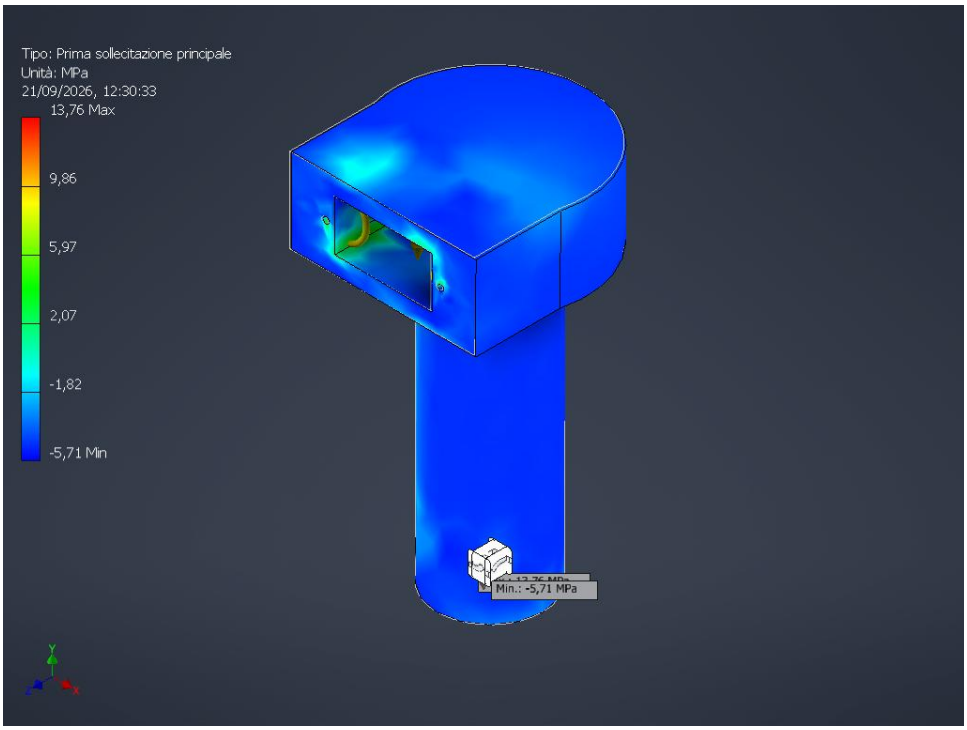
Figure

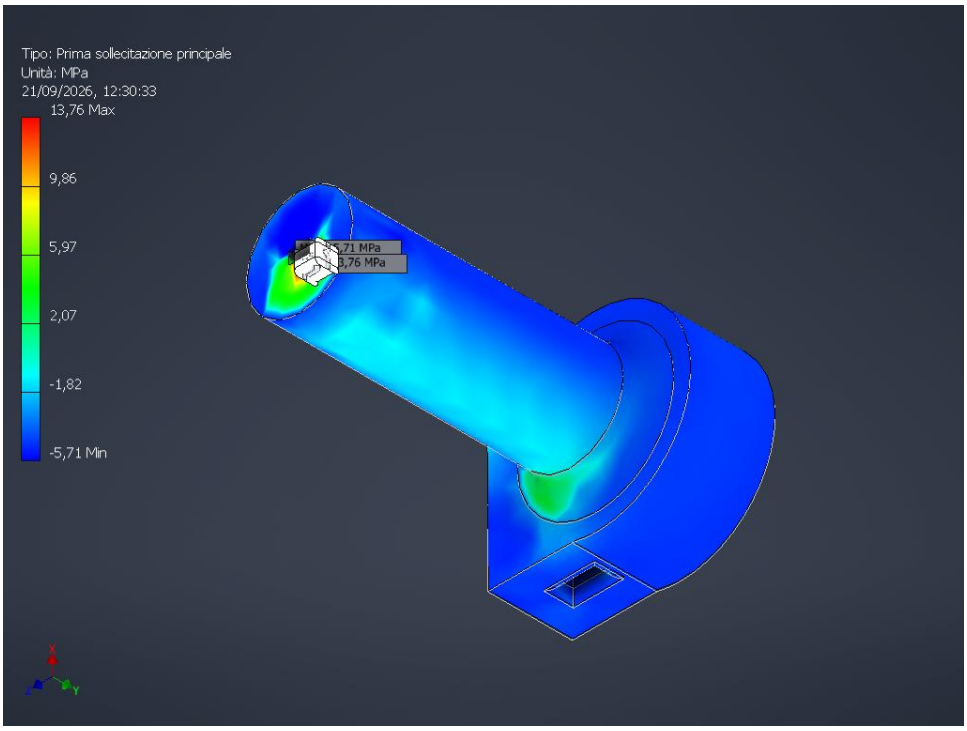
Sollecitazione di Von Mises



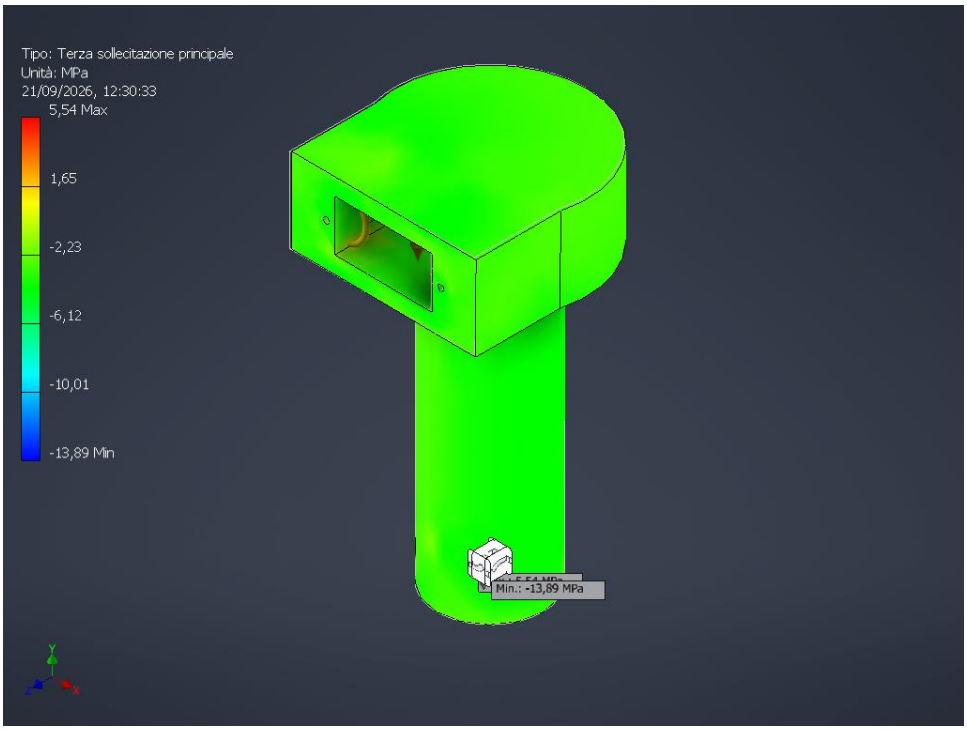


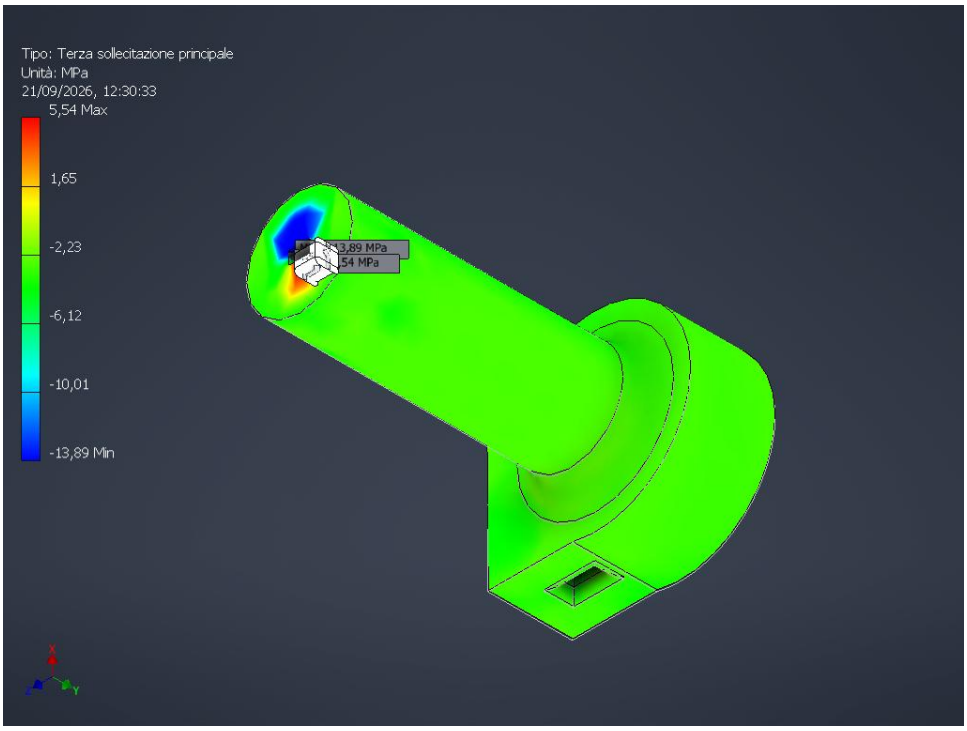
Prima sollecitazione principale



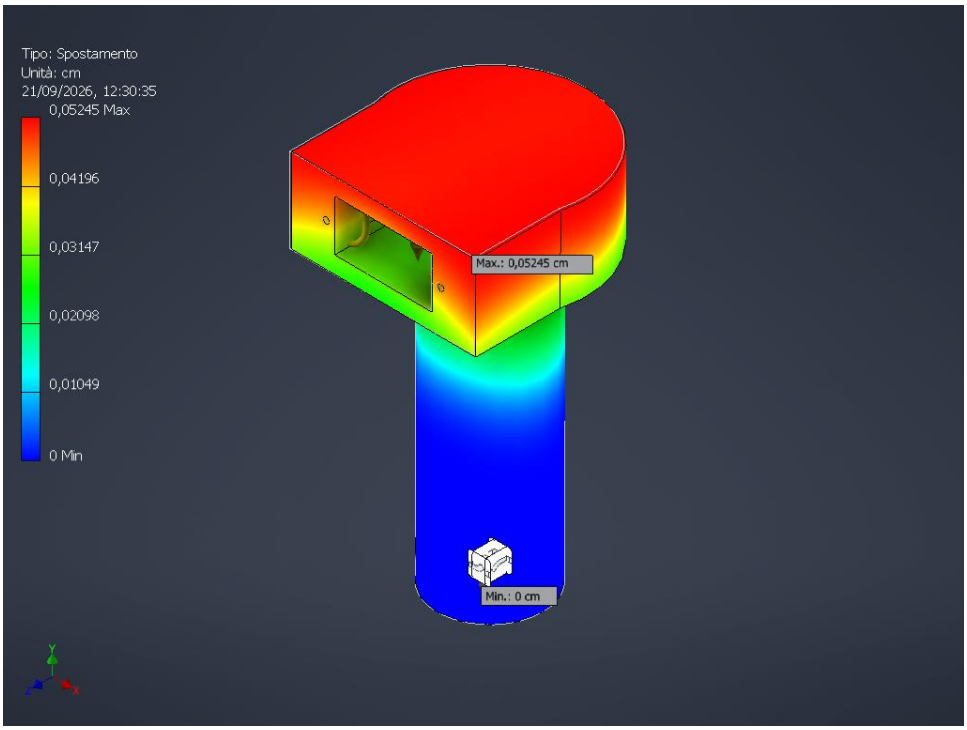


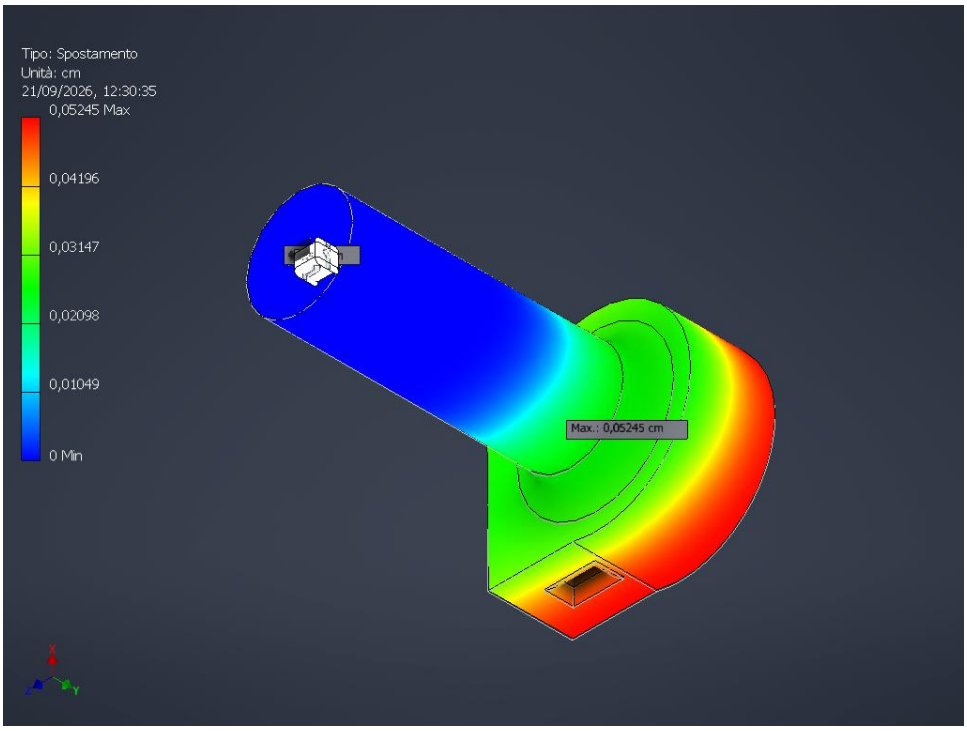
Terza sollecitazione principale



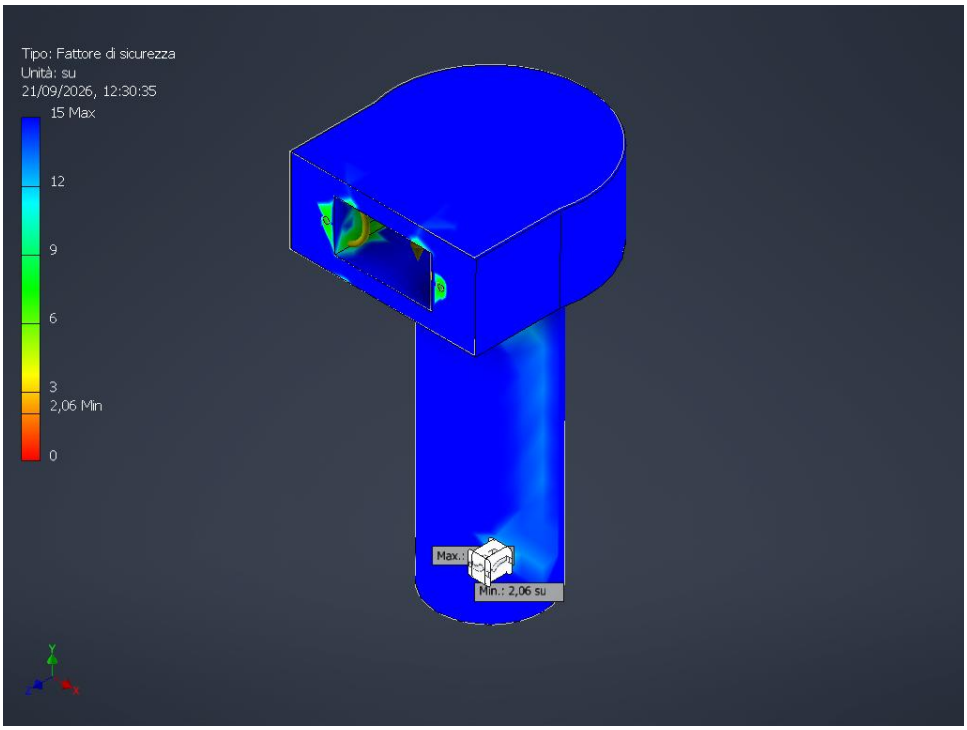


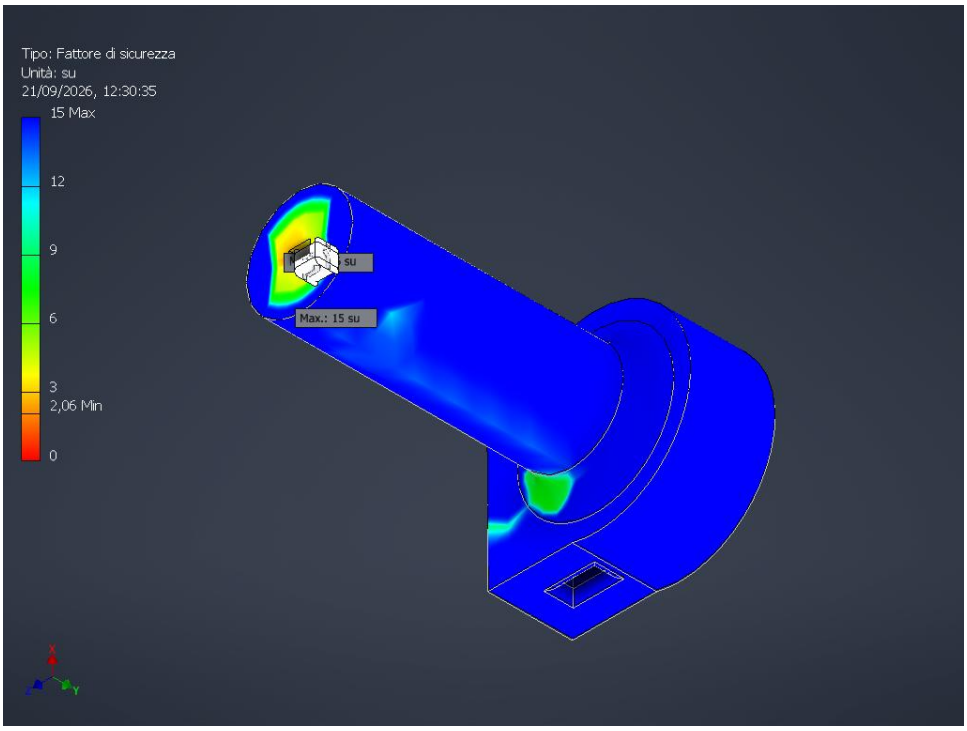
Spostamento



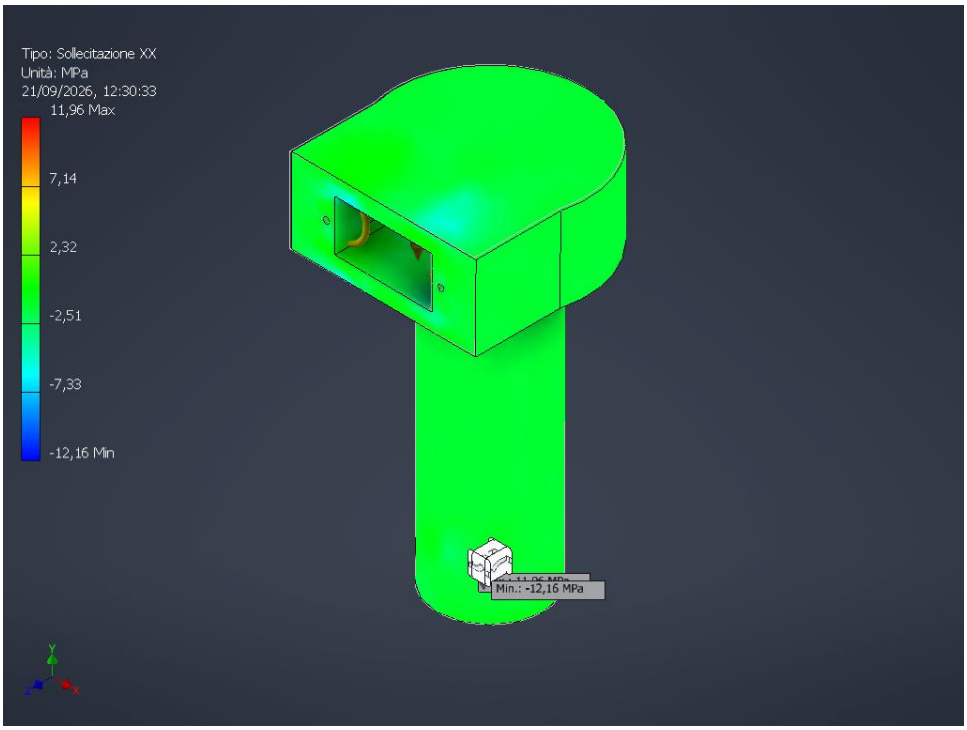


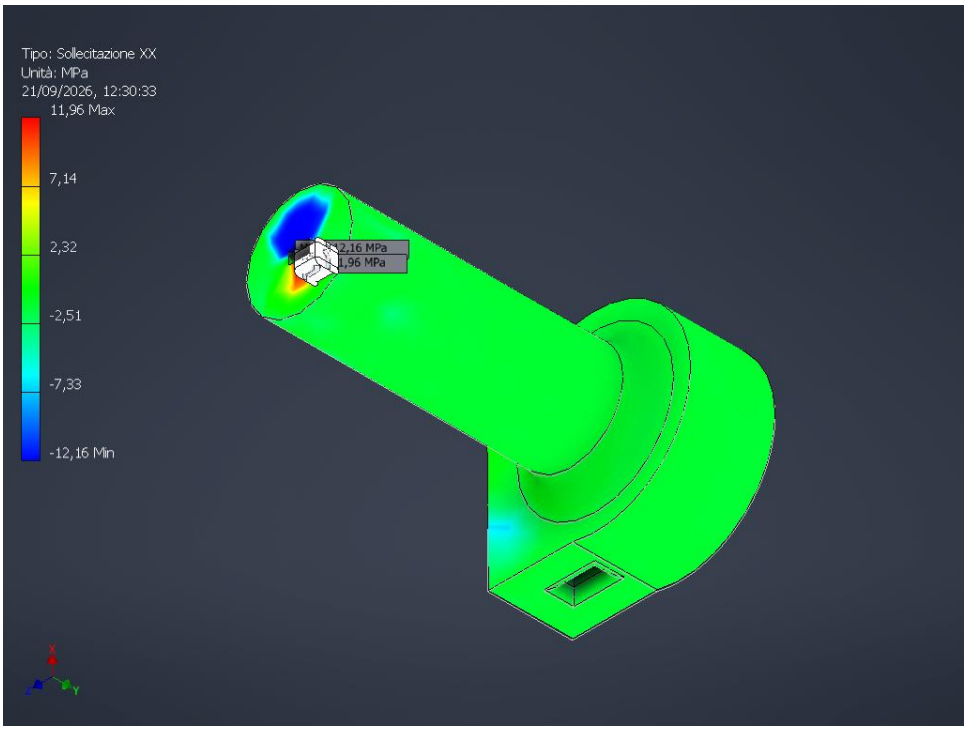
Fattore di sicurezza



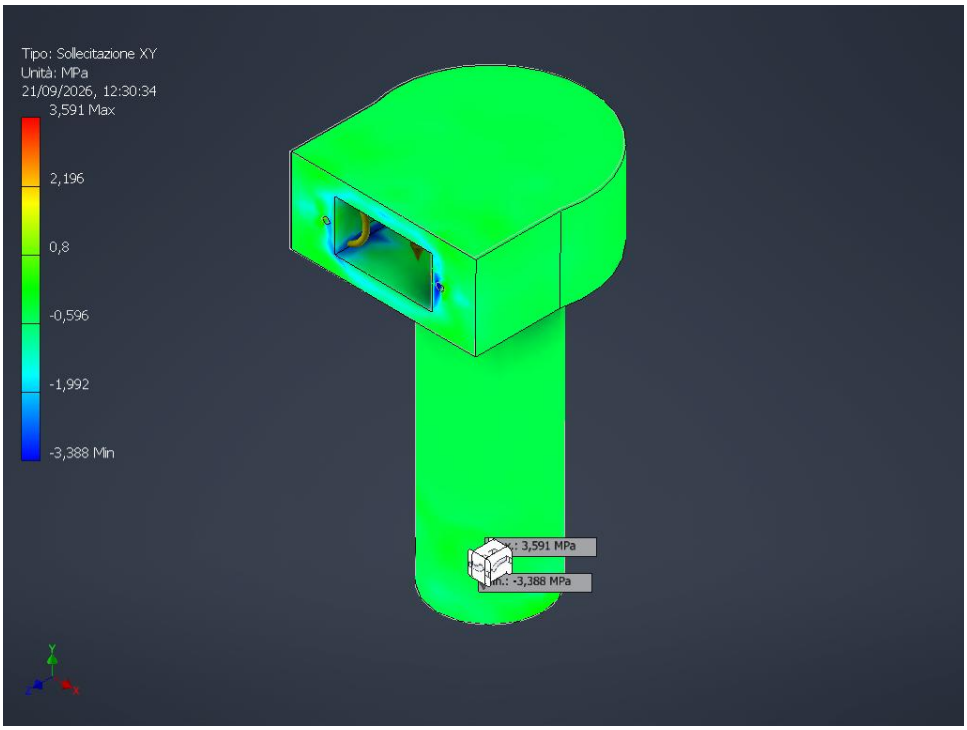


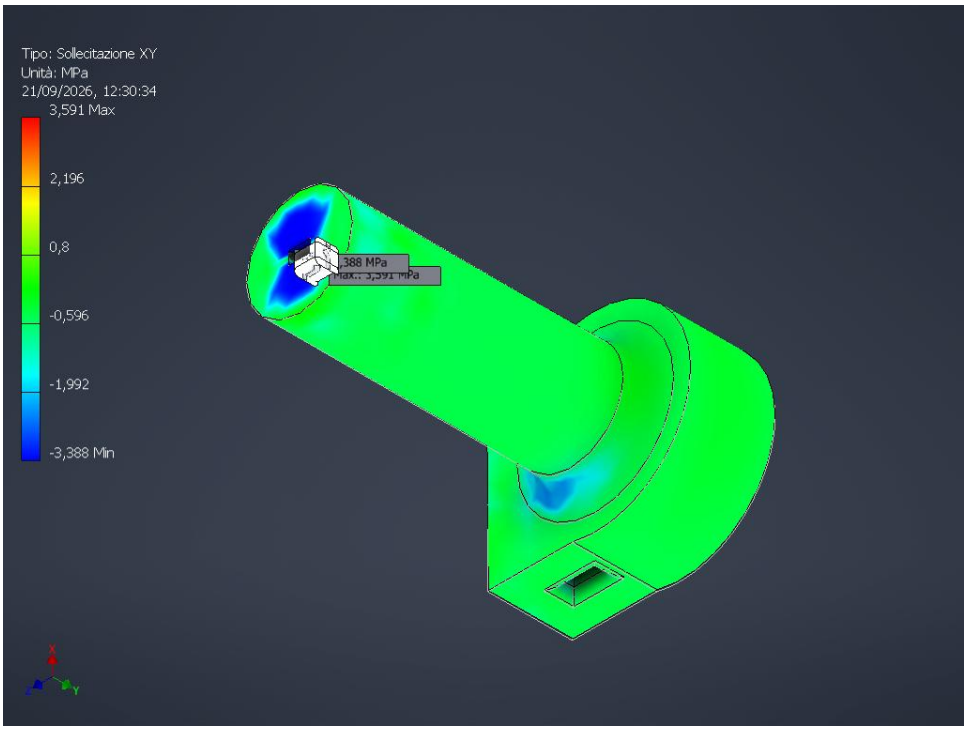
Sollecitazione XX



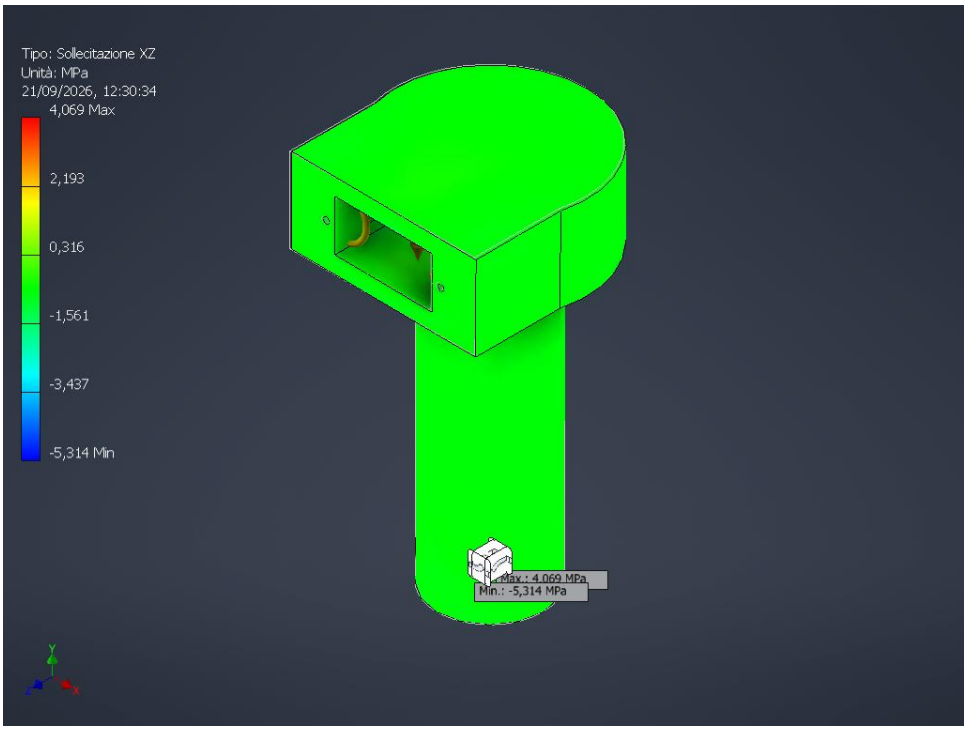


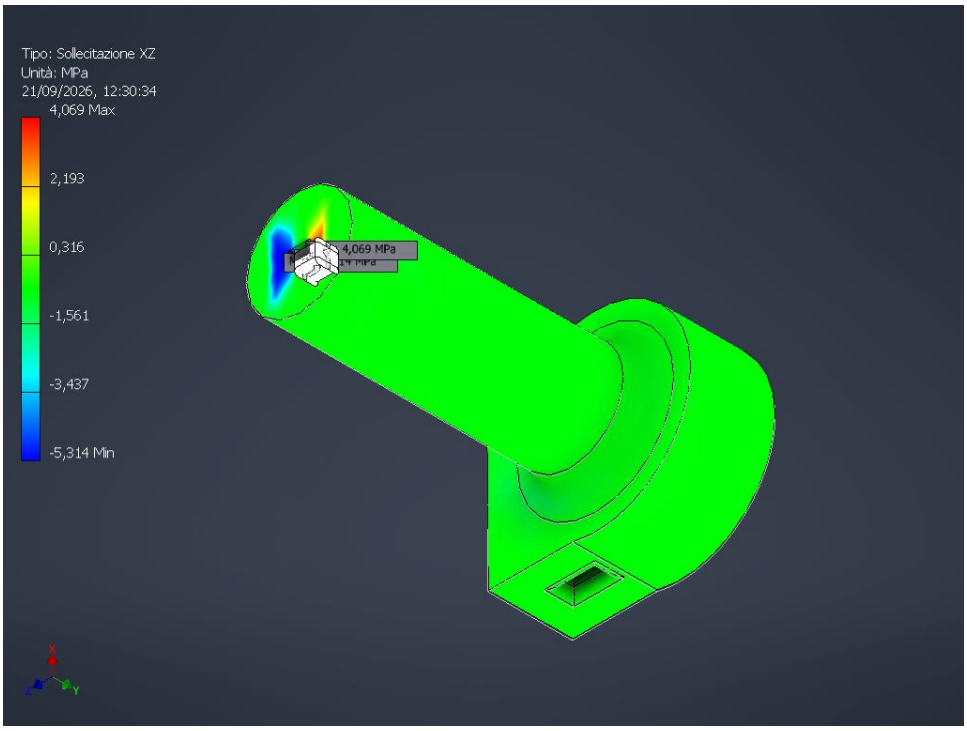
Sollecitazione XY



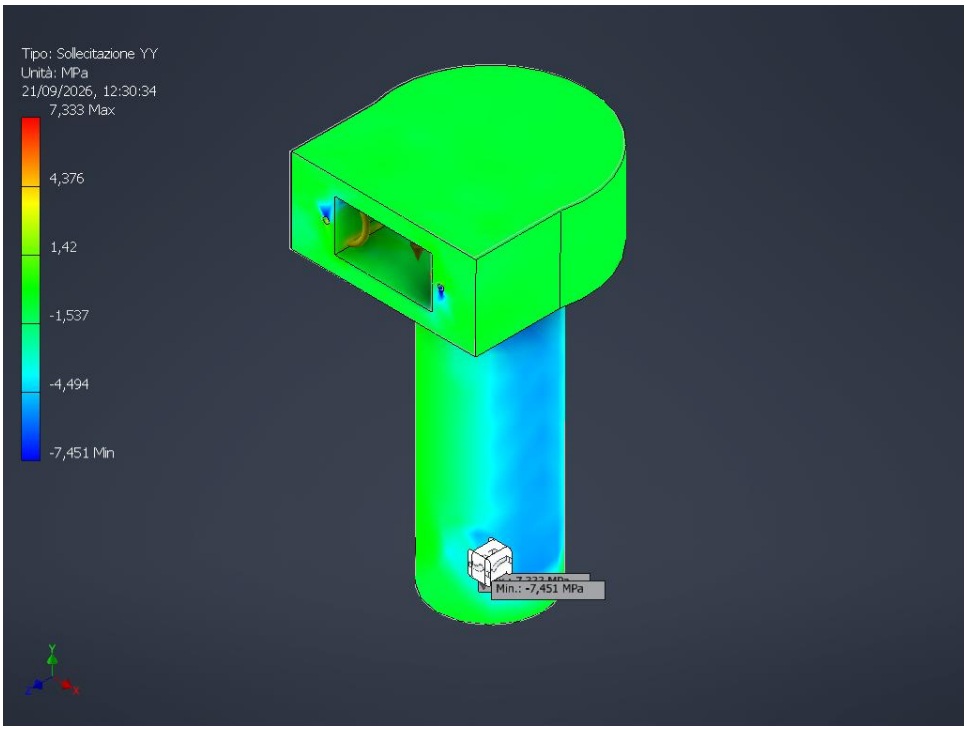


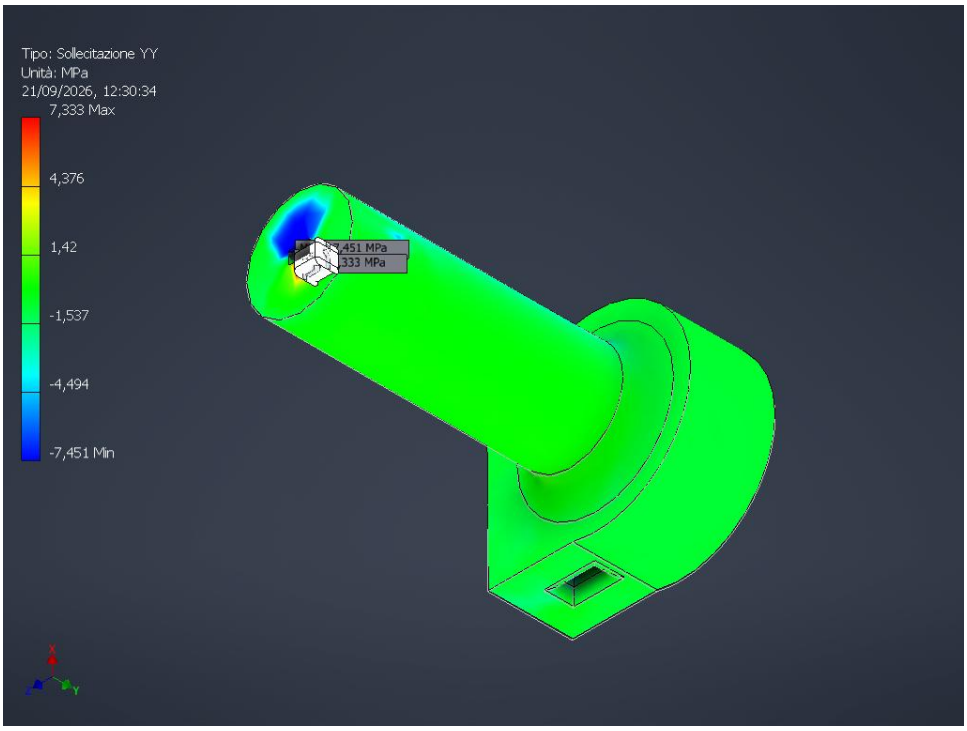
Sollecitazione XZ



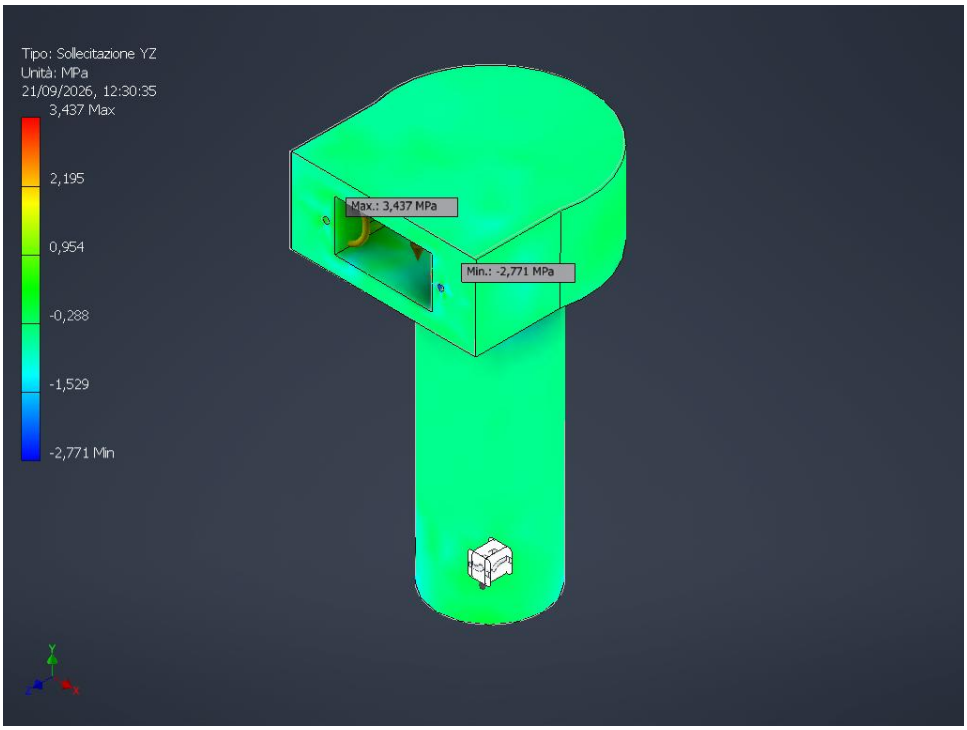


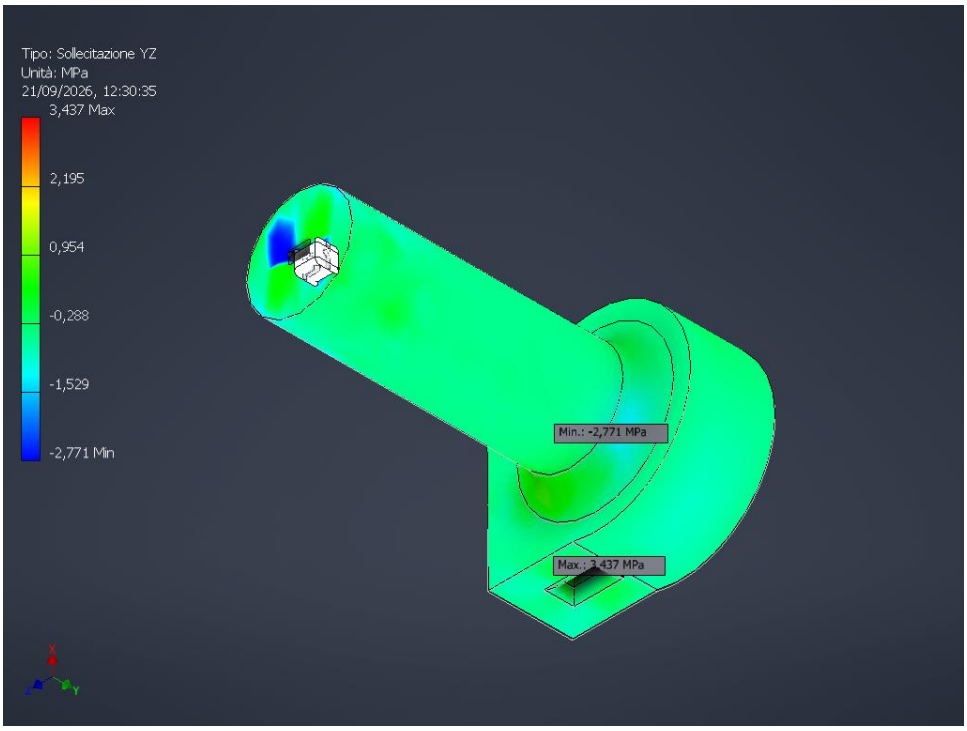
Sollecitazione YY



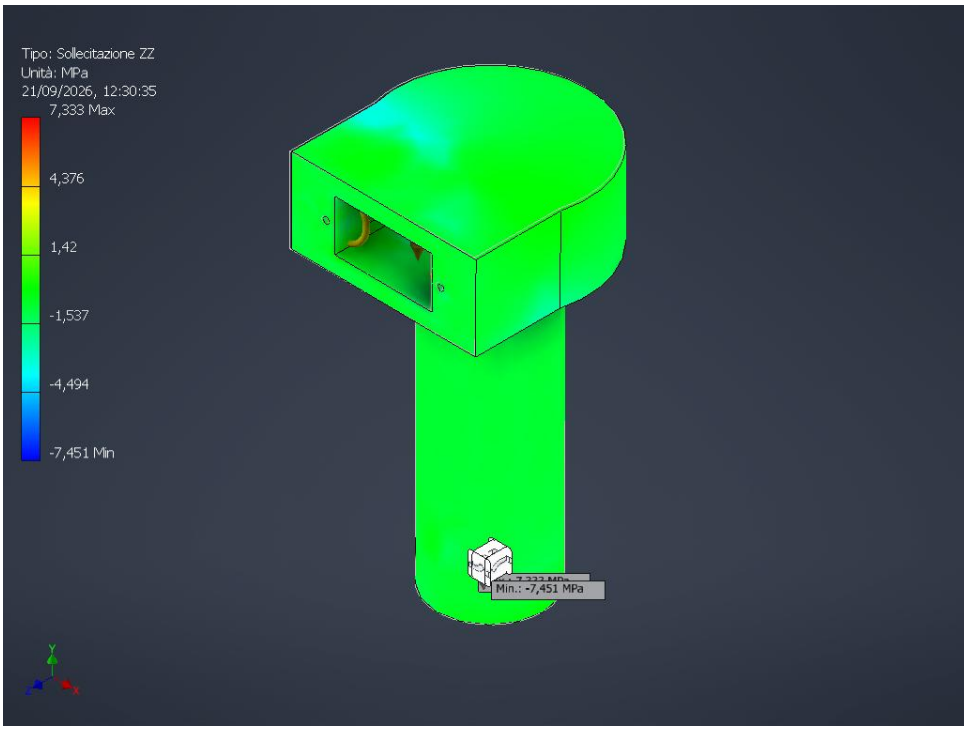


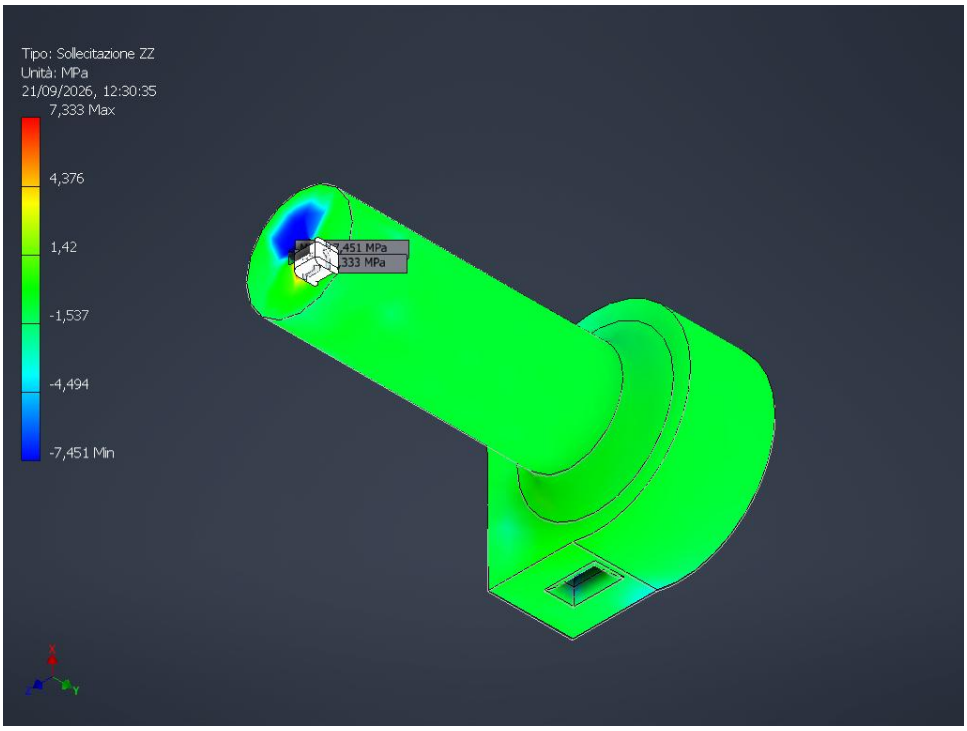
Sollecitazione YZ



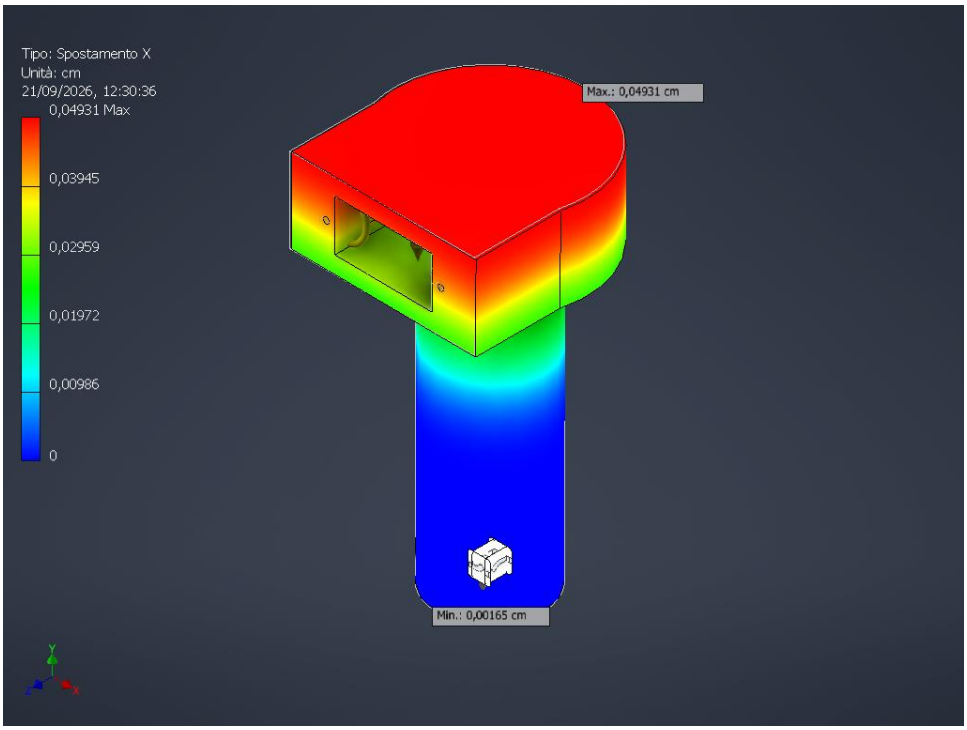


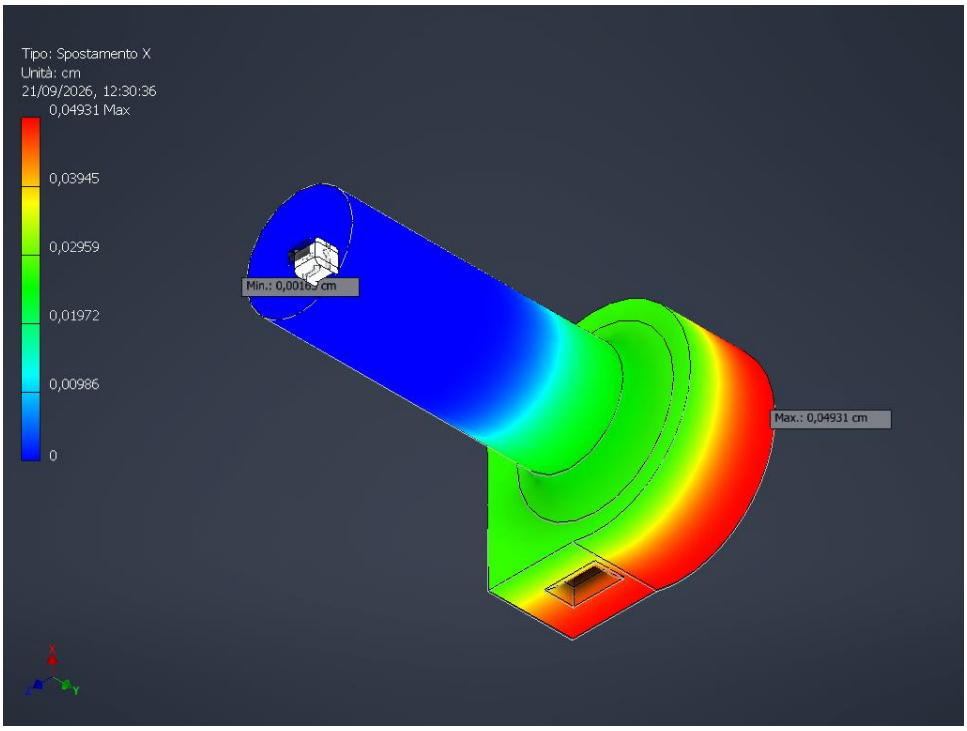
Sollecitazione ZZ



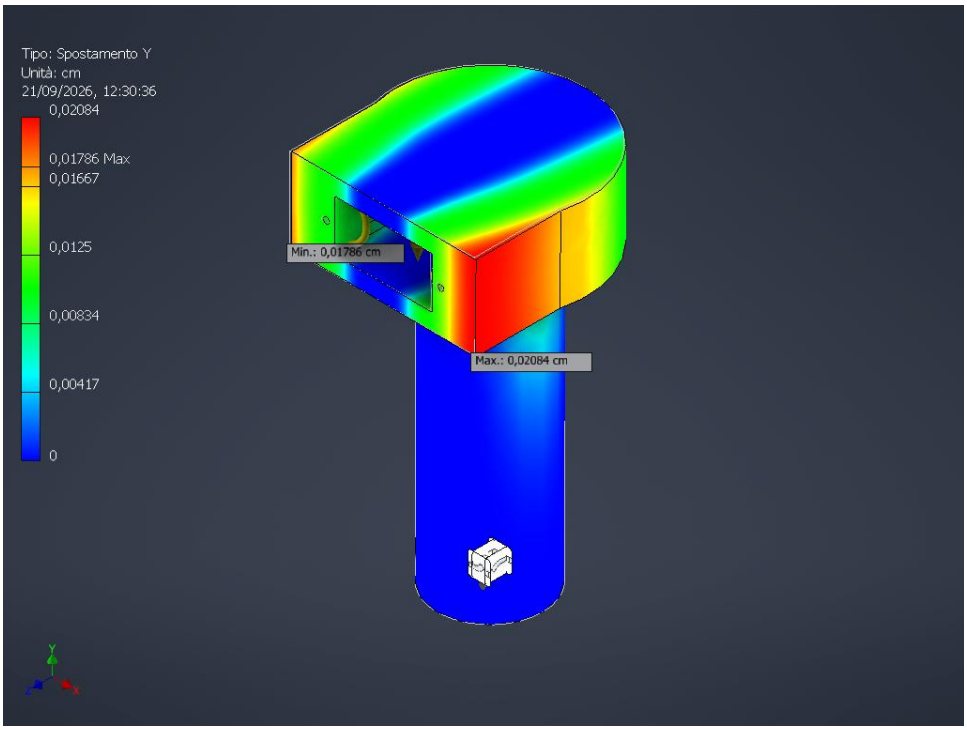


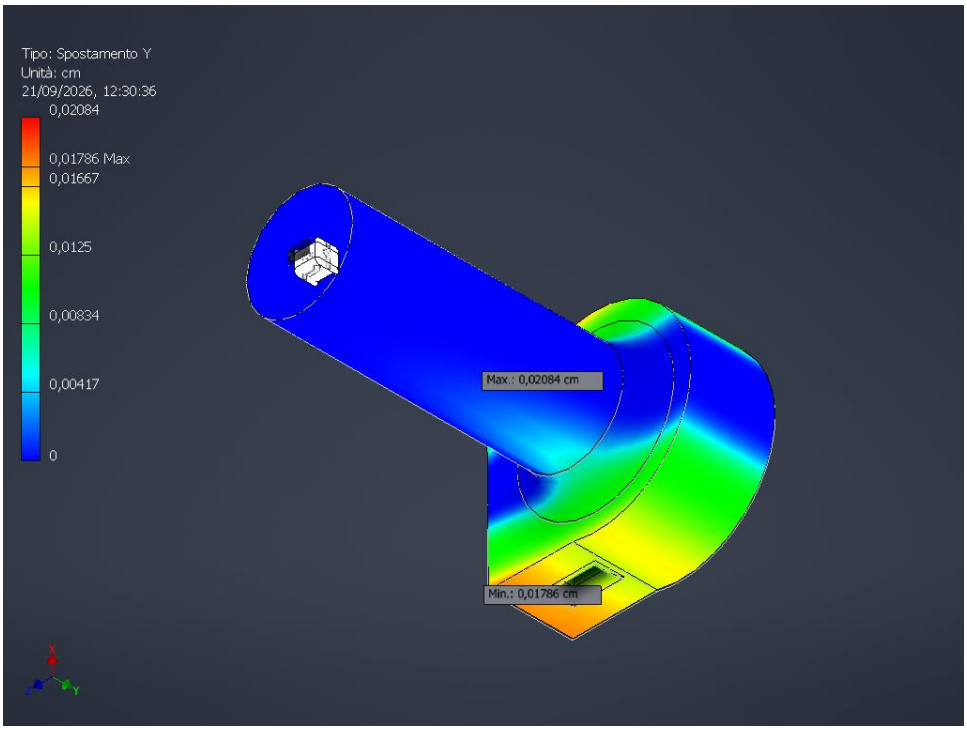
Spostamento X



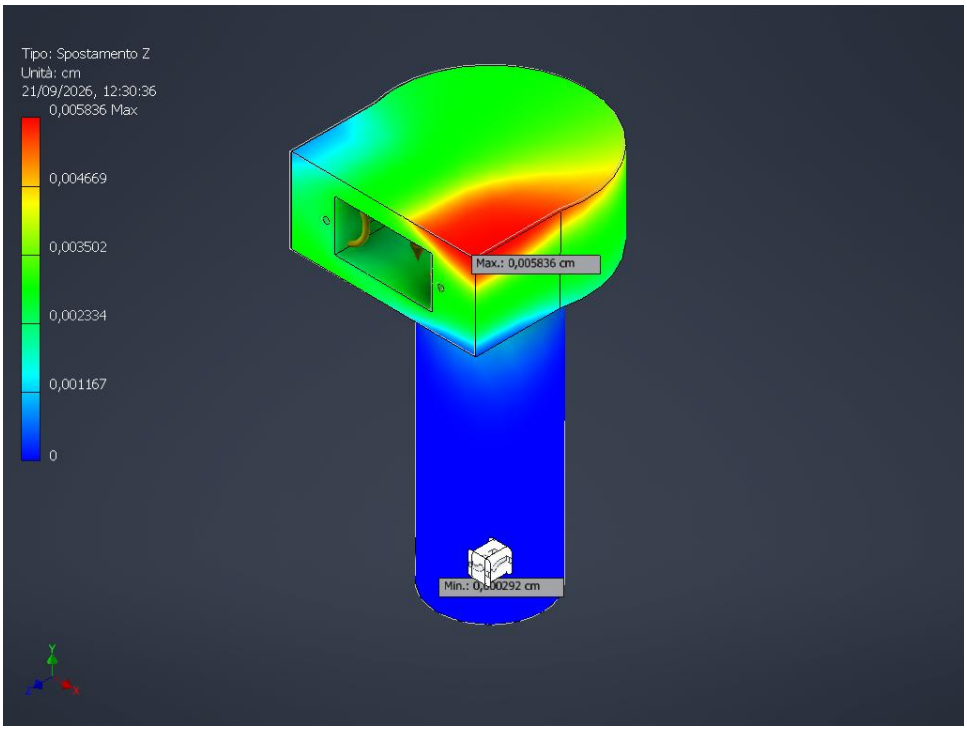


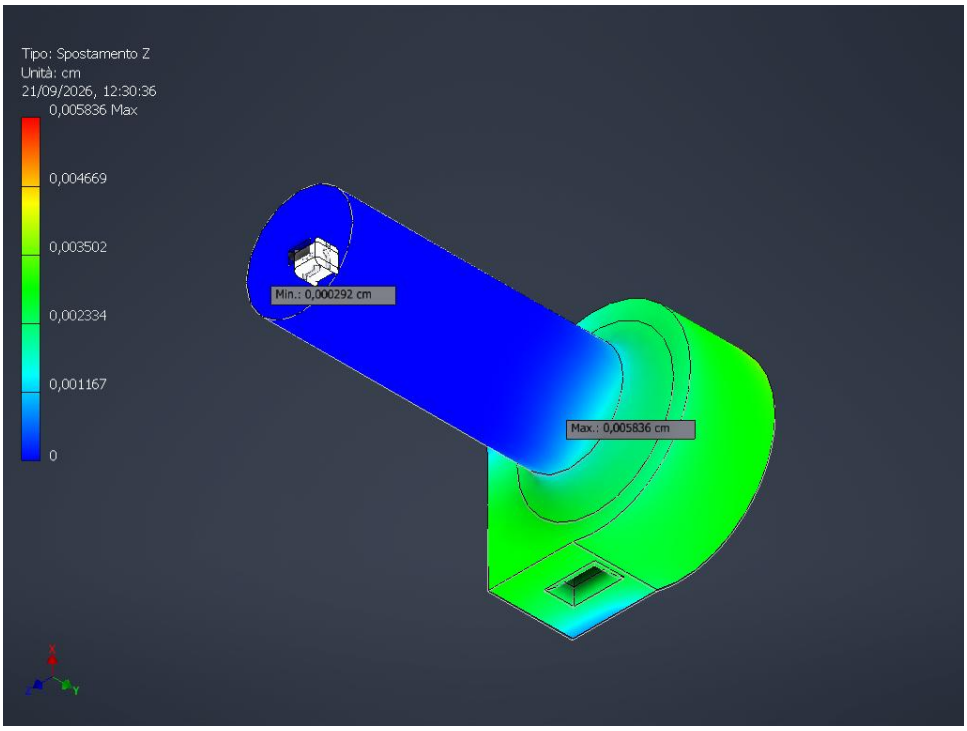
Spostamento Y



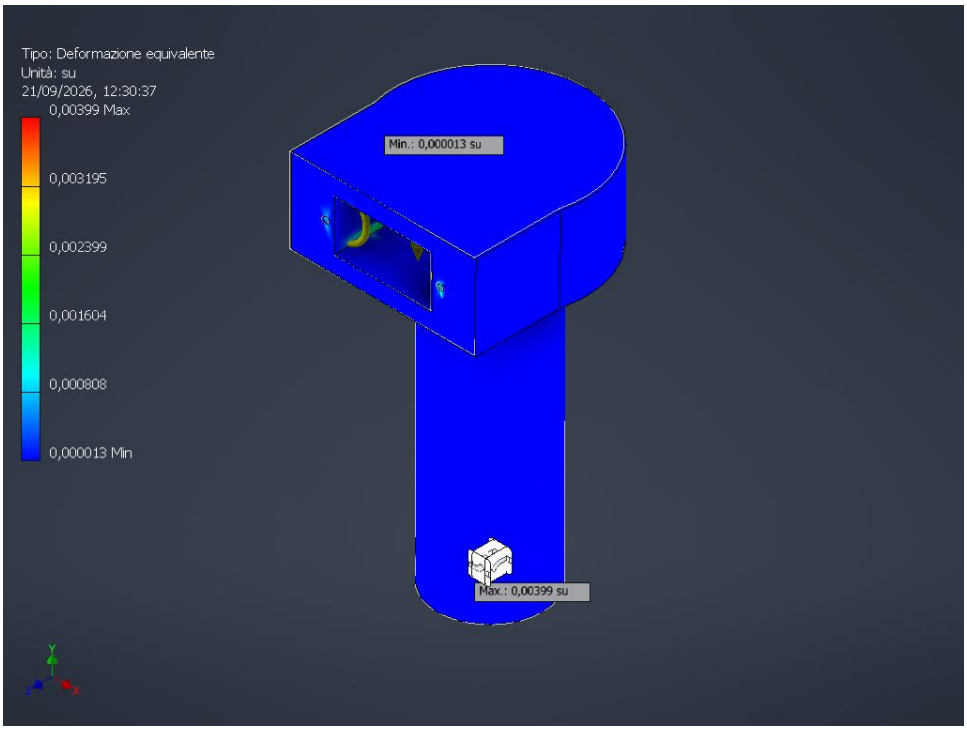


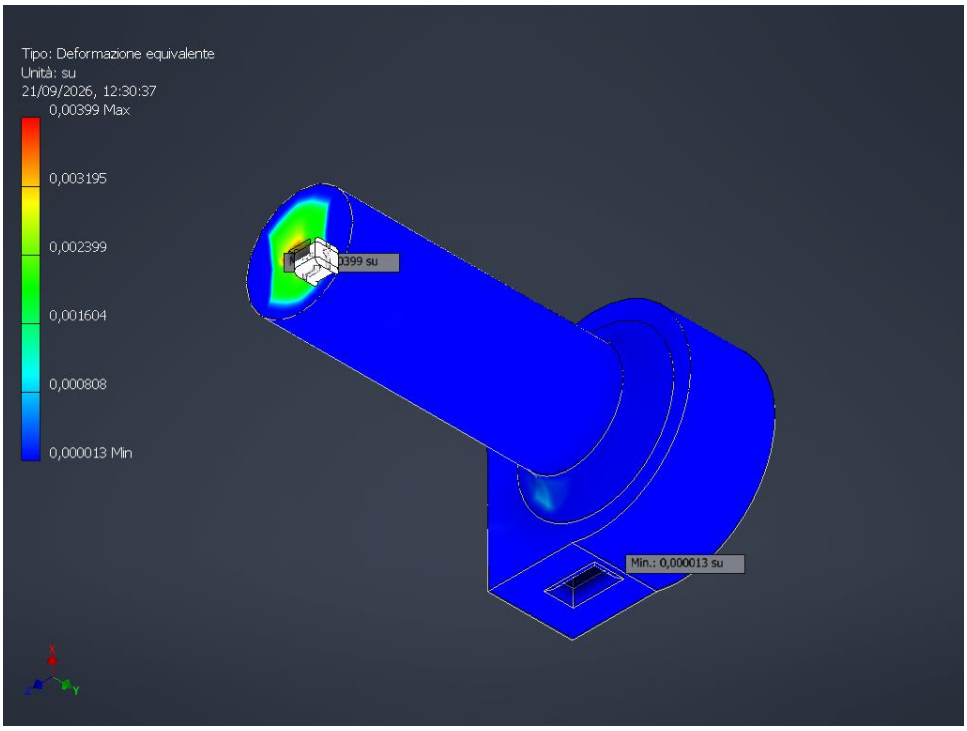
Spostamento Z



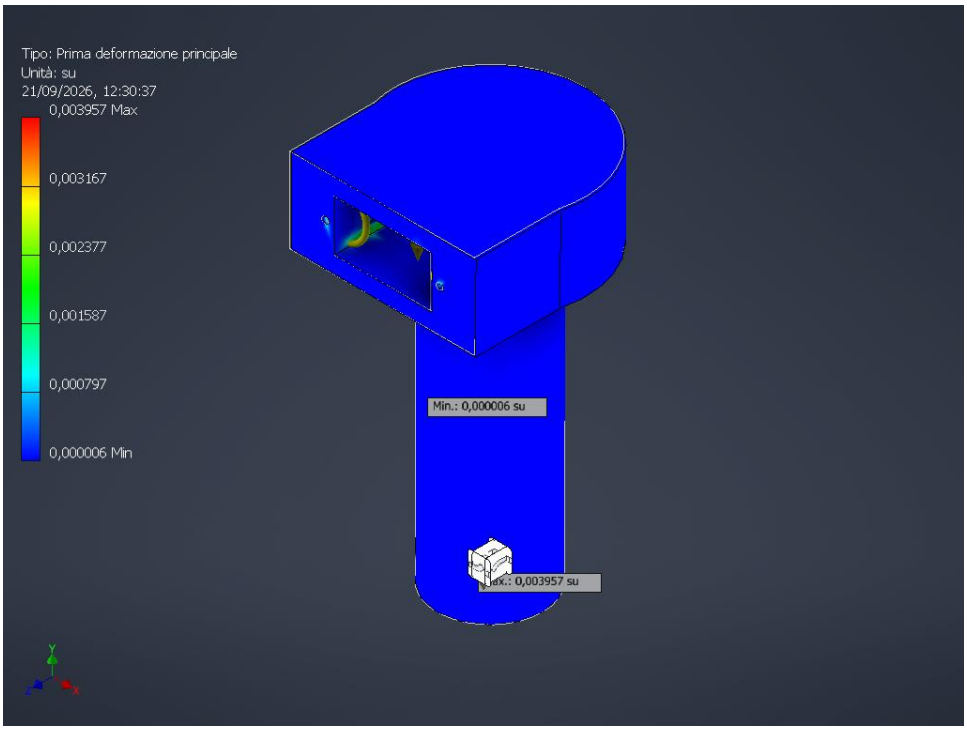


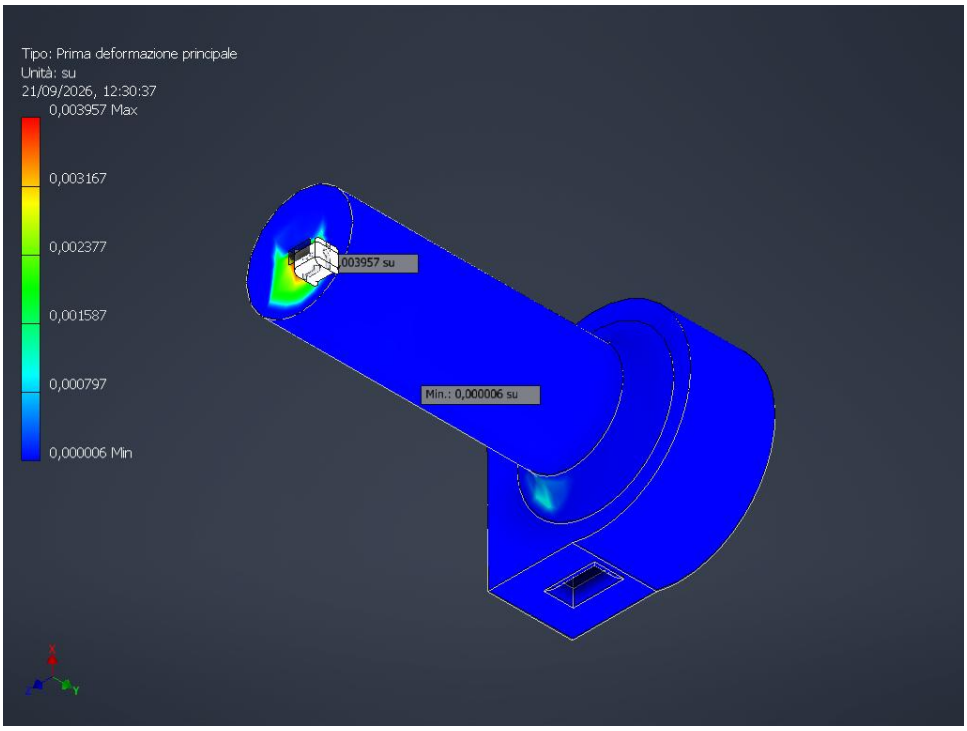
Deformazione equivalente



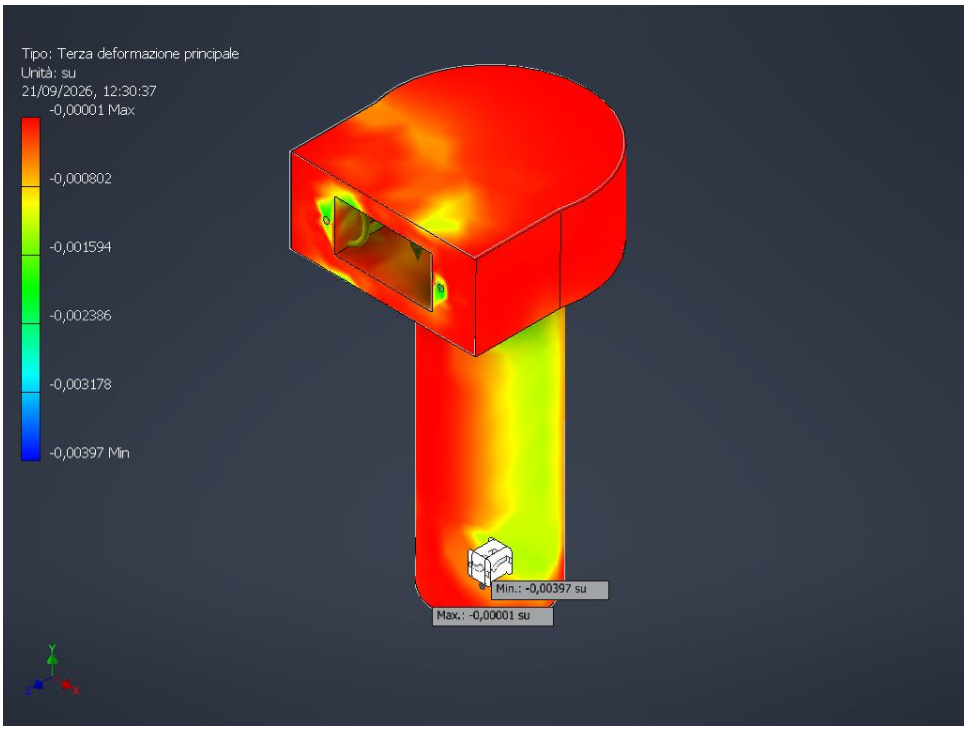


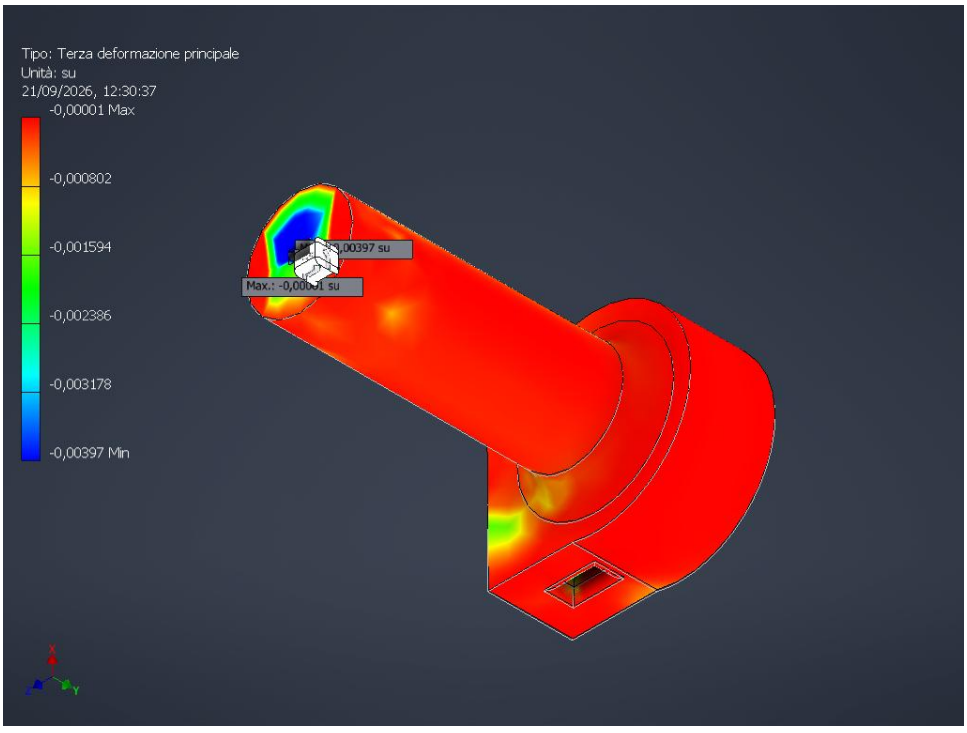
Prima deformazione principale



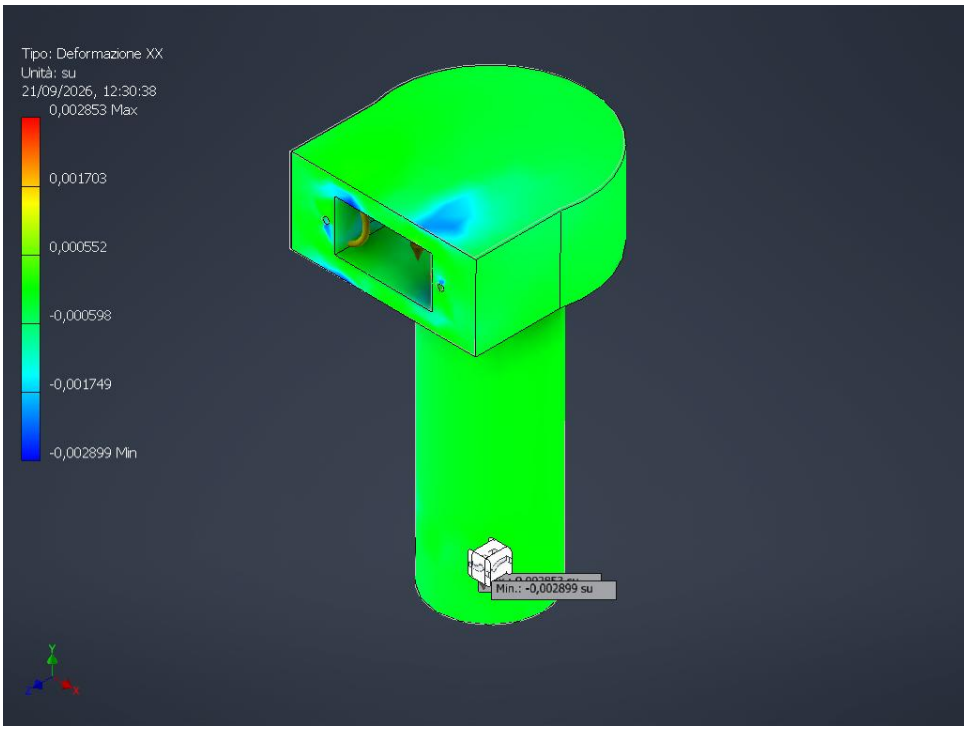


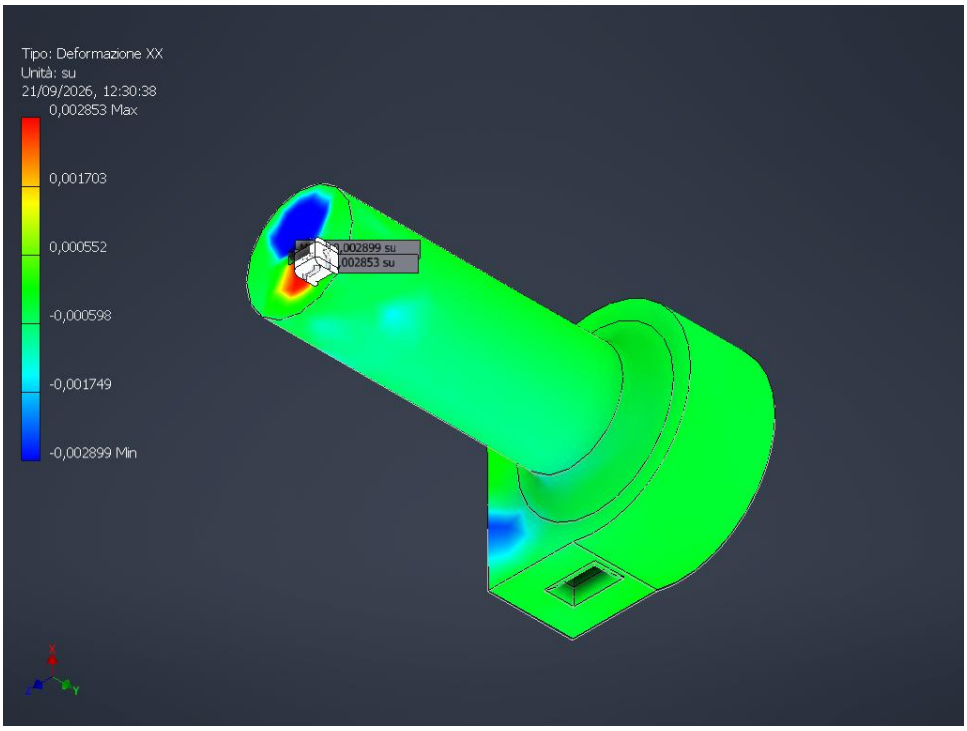
Terza deformazione principale



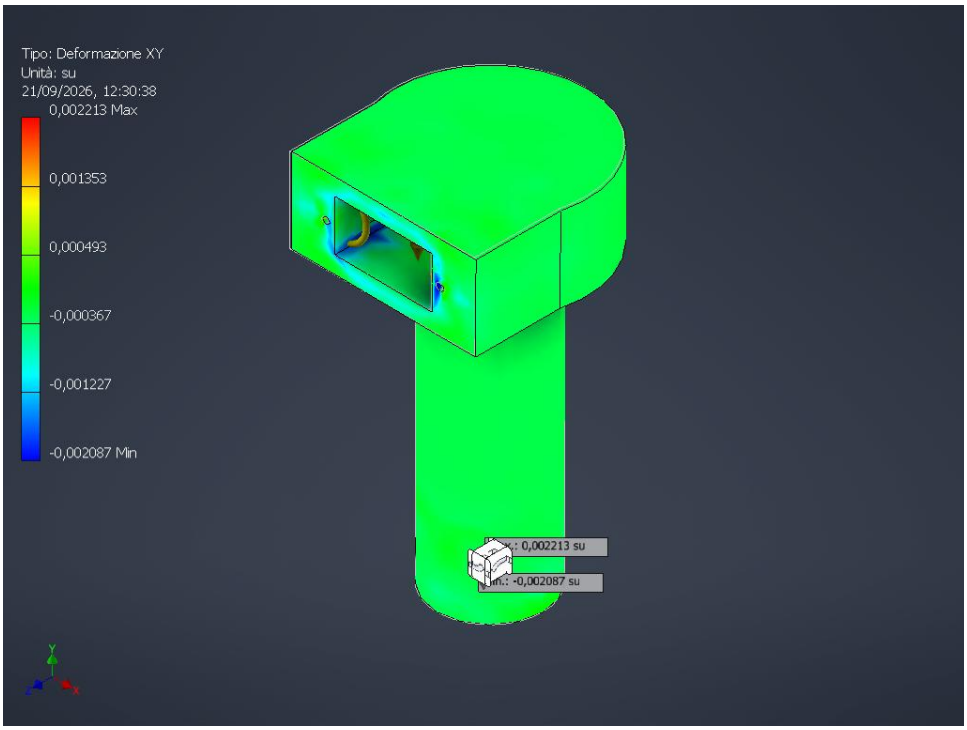


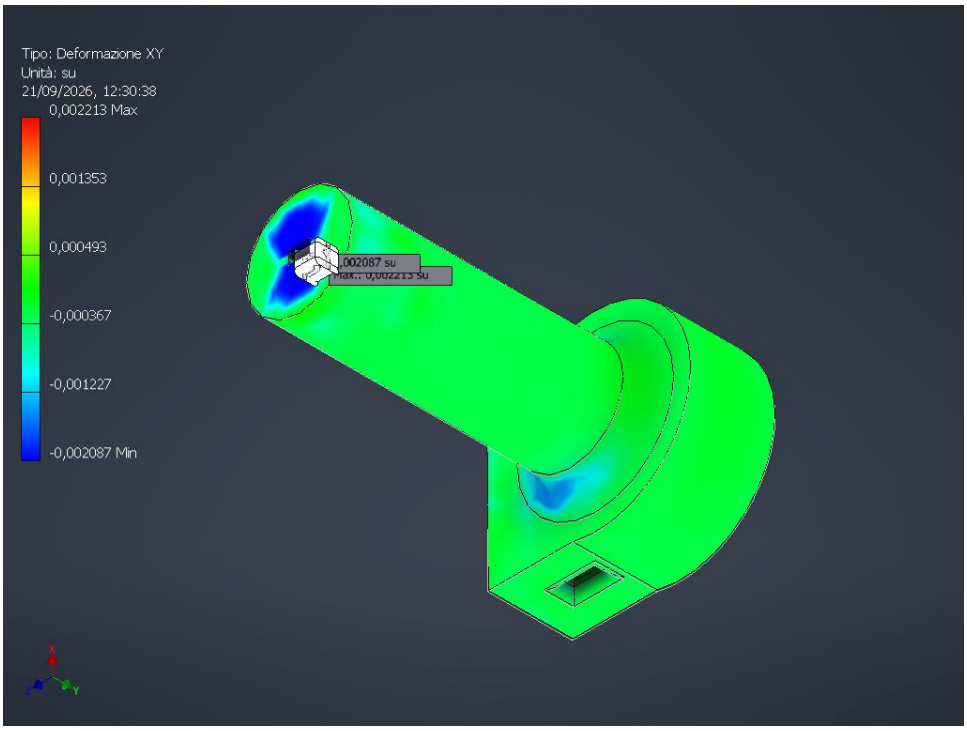
Deformazione XX



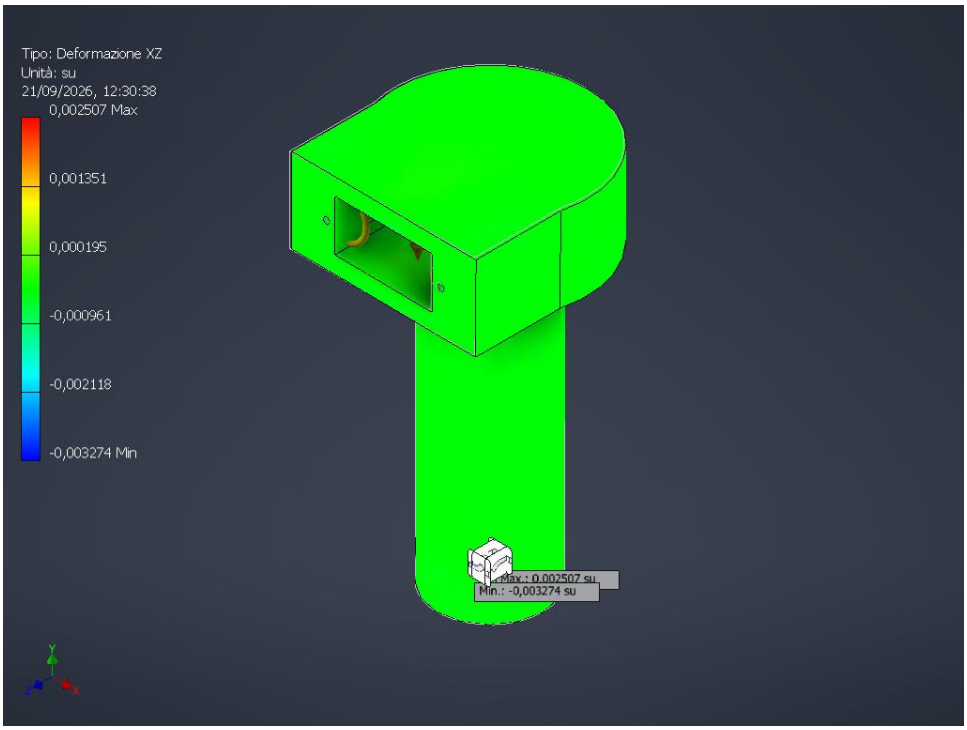


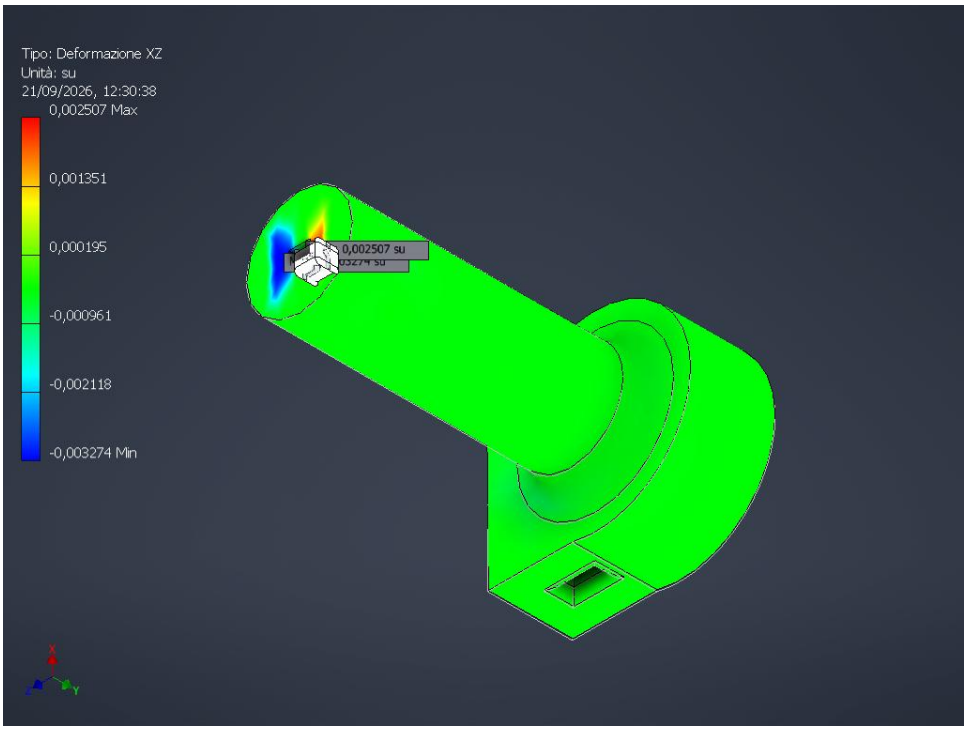
Deformazione XY



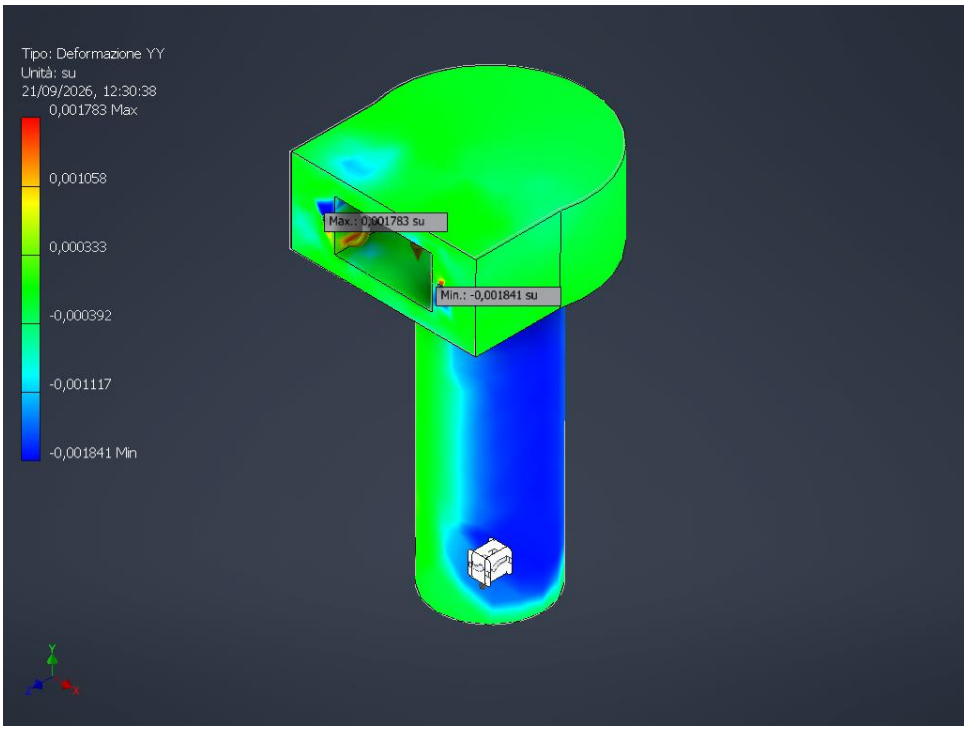


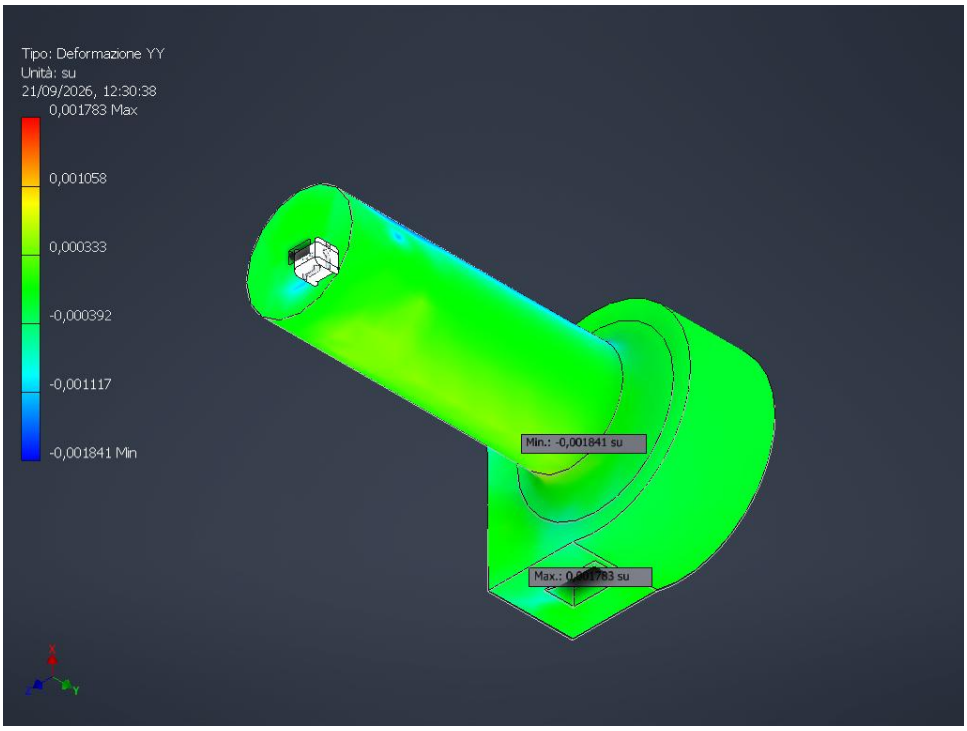
Deformazione XZ



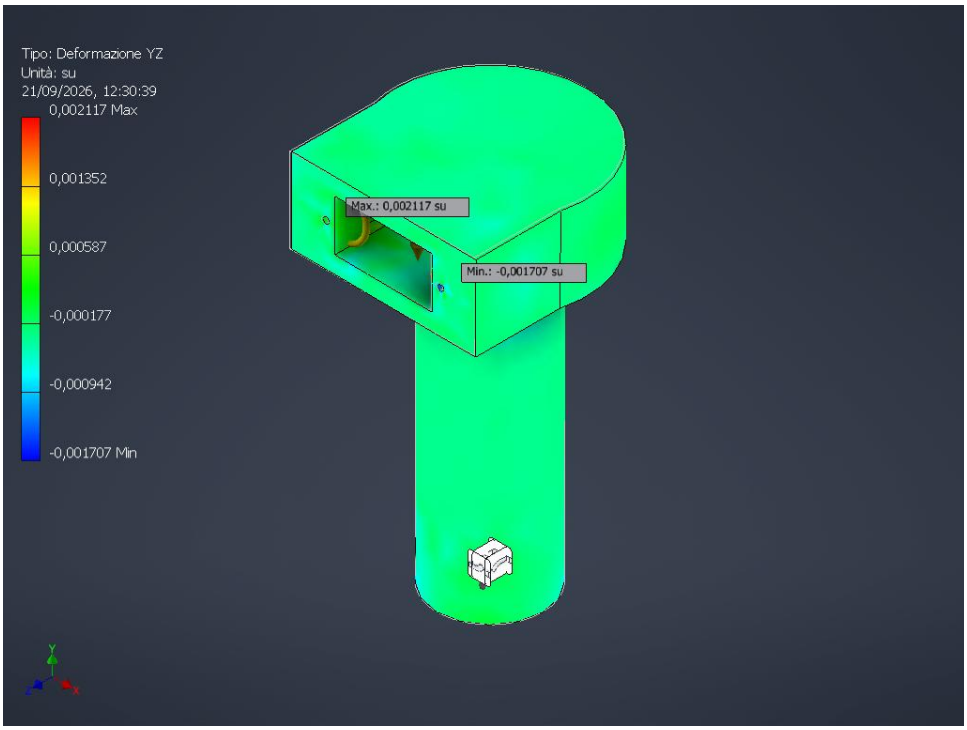


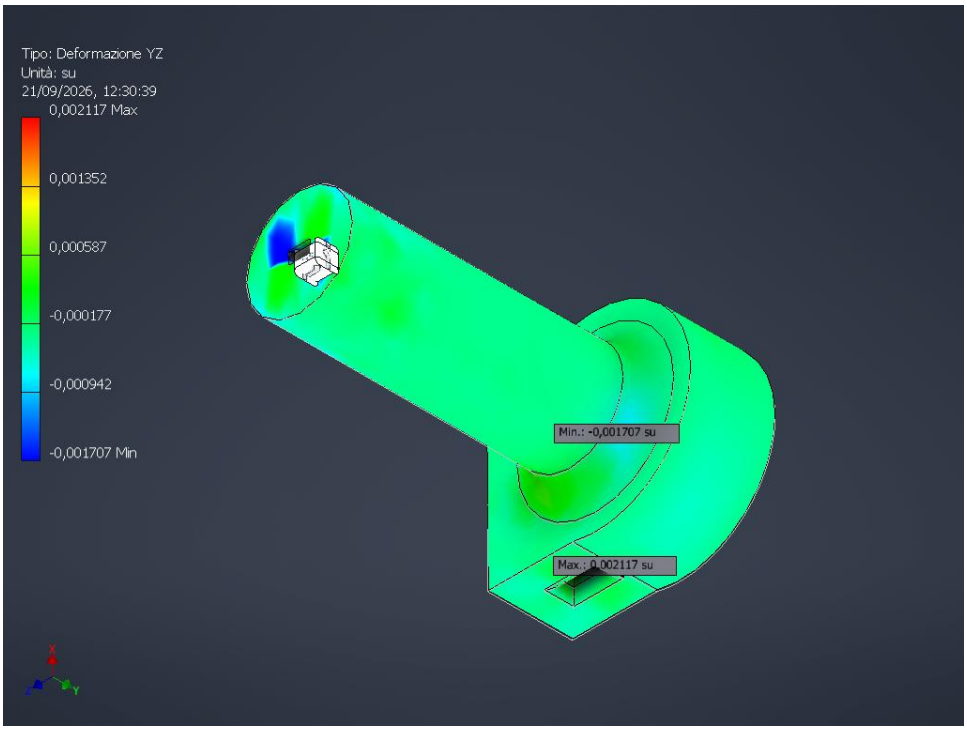
Deformazione YY



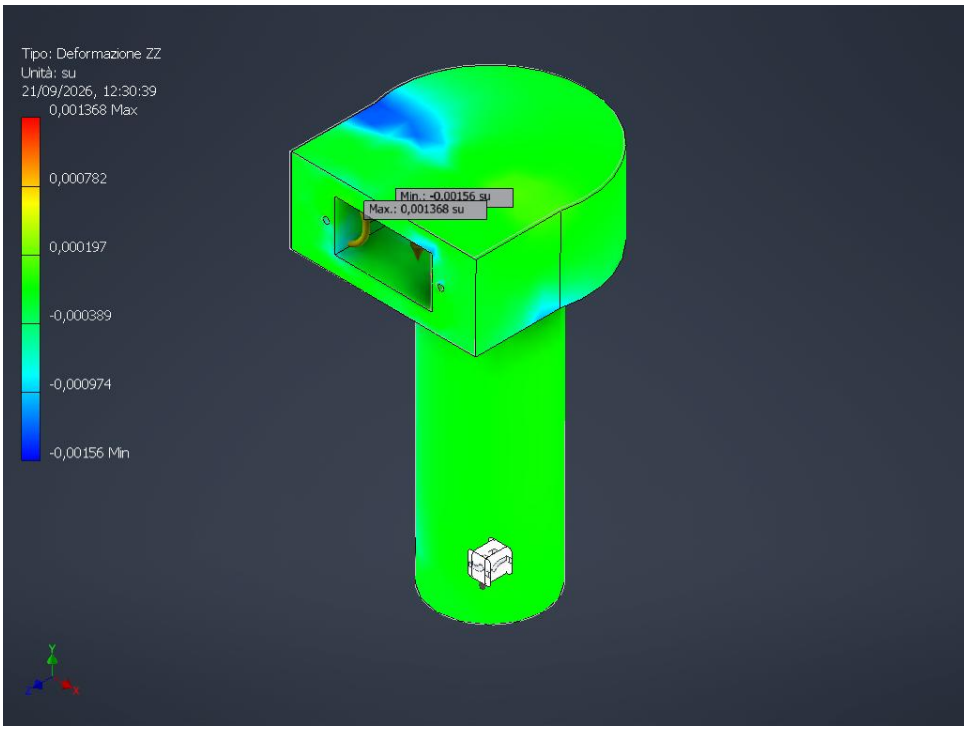


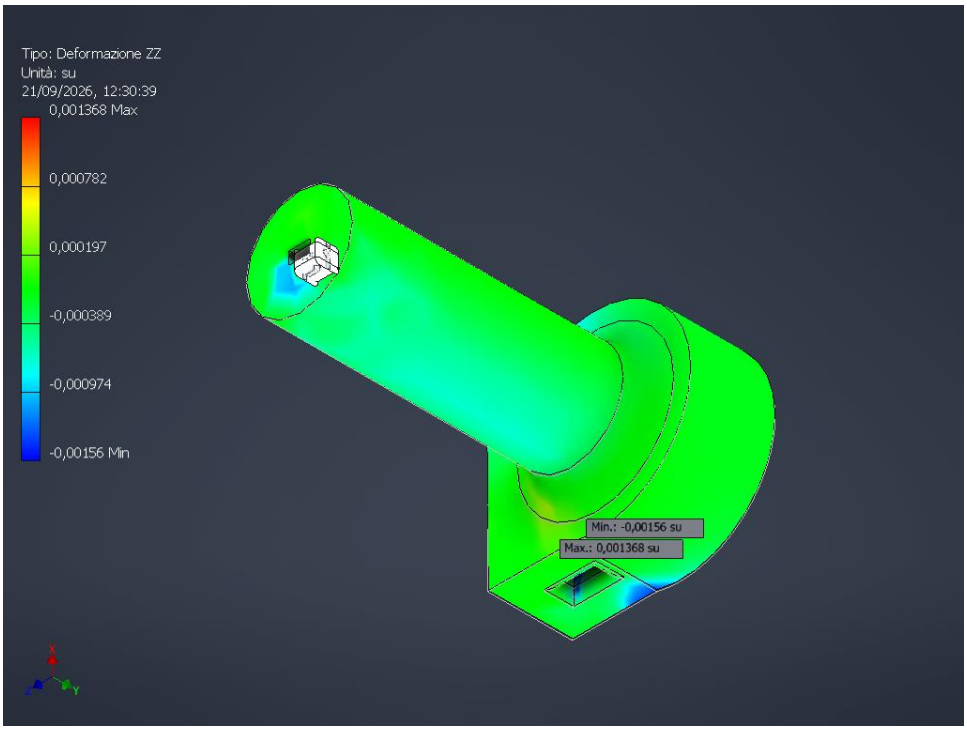
Deformazione YZ





Deformazione ZZ





C:\Users\nuzzo\Desktop\Braccio 4Dof\360° Rotating arm v2\360° Rotating arm v2.ipt